

PROJECTION-TRANSPORT STEERING: CONDITIONAL CONTROL OF REASONING BEHAVIOR IN LLMs

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Paper under double-blind review

ABSTRACT

Activation steering controls the behavior of a language model at inference time by editing its residual stream, without touching the weights. The edits in use are global: one vector is added at every token, so an edit strong enough to remove a behavior reaches the inputs on which it was never present. In this paper we propose Projection-Transport Steering (PTS), a framework that separates when to intervene from how to intervene. PTS takes the action to be the optimal transport map onto a target law in a low-dimensional behavioral subspace, and the decision to be a likelihood-ratio test on a second, detection axis. We show that this map is the unique minimum-displacement edit local to the subspace, that the selectivity of any decision rule is capped by how well its axis separates the subpopulations it must distinguish, and that gate and map are jointly optimal. Using PTS, one can remove most of a reasoning behavior while keeping three quarters of the answers that never exhibited it, and leave task accuracy, calibration, code synthesis and open-ended generation at their unsteered values. The operator can also abstain: carried without refitting to a domain that carries six instances of the behavior in three hundred, it is the exact identity, at two dot products and a table lookup per token. Our experiments across two model scales and three reasoning benchmarks show that a behavior edit can be made local in the space it acts on and conditional on the input it acts for.

1 INTRODUCTION

A striking feature of large language models is that high-level behaviors are carried by directions in their activations. This has given rise to activation steering (Turner et al., 2023; Panickssery et al., 2024; Zou et al., 2023; Li et al., 2023), a family of inference-time interventions that read such a direction off contrastive examples and add it back to the residual stream, changing how a model refuses, hedges, or writes, at the cost of one vector addition per token and no retraining at all.

Problem: a steering edit cannot tell where the behavior is. The edit these methods apply is the same on every input. A refusal vector added at a useful strength raises refusal on harmful and harmless prompts alike (Arditi et al., 2024), and the difficulty compounds when the target is a behavior the model shows only sometimes. Consider overconfidence in multiple-choice reasoning. The states a practitioner wants to separate are two *confident* ones, wrong and right, and the dose that removes the first also removes the second: in our measurements the additive edit that eliminates every overconfident-wrong answer eliminates every confident-right answer with it, and takes code synthesis and fluent generation down with them (Figure 1, left). Tuning the coefficient trades one failure for the other rather than escaping either, because a constant shift moves the bulk of the distribution exactly as far as it moves the tail.

Contribution: separating when to intervene from how. We introduce Projection-Transport Steering, a framework in which an edit is two independent objects. The *action* is a transfer function on the projection of the residual stream onto a behavioral subspace, and we ask which transfer function is cost-minimal instead of fixing it by hand. The *decision* is a test on a second axis, fitted to tell the two confident states apart, that determines whether the action runs at all. Table 1 places the published methods in this frame: they occupy the top rows, differing only in the shape of the transfer function,

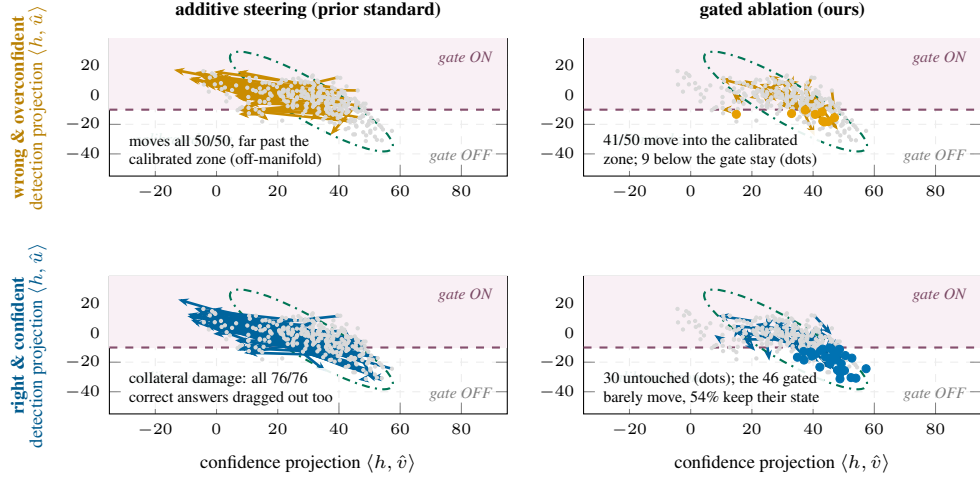


Figure 1: Each dot is one reasoning trace before steering, an arrow is its measured movement after steering on real rollouts, and a bold dot without an arrow was left untouched. The horizontal axis is the confidence projection $\langle h, \hat{v} \rangle$ and the vertical axis the detection projection $\langle h, \hat{u} \rangle$. The pink band is gate ON and the green ellipse is the 2σ contour of the calibrated extraction-split states (Appendix N). Additive steering (left) overshoots all 50 OCW far past the calibrated zone and drags all 76 CR along. Gated ablation (right) moves OCW into the zone and leaves 30 CR bitwise untouched.

method	transfer function $g(p)$	decides when	cost / token
ActAdd, CAA (Turner et al., 2023; Panickssery et al., 2024)	$p + \alpha$	—	$\Theta(d)$
ITI (Li et al., 2023)	$p + \alpha$ on probe-selected heads	—	$\Theta(d)$
RepE (Zou et al., 2023)	$p + \alpha(x)$	—	$\Theta(d)$
directional ablation (Arditi et al., 2024)	0	—	$\Theta(d)$
SAE clamping (Templeton et al., 2024)	c	—	$\Theta(d d_{\text{sae}})$
CAST (Lee et al., 2024)	$p + \alpha \mathbf{1}[p > \tau]$	prompt similarity	$\Theta(d)$
MiMiC (Singh et al., 2024)	full-space affine map	—	$\Theta(d^2)$
Linear-ActT (Rodríguez et al., 2025a)	$(1-\lambda)p + \lambda(\omega p + \beta)$, per neuron	—	$\Theta(d)$
PTS (ours)	$S(p)$, the transport map to ν	Neyman–Pearson gate	$\Theta(kd)$

Table 1: **Linear activation steering is one family.** Every operator in this literature is a transfer function g applied to a k -dimensional projection of the residual stream, with $k=1$ on a behavioral direction for the top rows, $k=d_{\text{sae}}$ for SAE clamping and $k=d$ for the full-space maps. The published methods differ only in the shape of g , in the k it acts on, and in whether anything decides when to apply it. Theorem 1 identifies the cost-minimal g for a given target law, Cor. 1 says which row is optimal and when, and Thm. 2 bounds what any *decides when* column can buy.

and the column that decides when to act is almost empty. Figure 1 shows what filling it buys on real reasoning traces.

Steering is optimal transport. Once the subspace and the target law are fixed, there is nothing left to design. In Section 3 we prove that the optimal transport map between the projection laws is the unique minimal-perturbation intervention local to that subspace, and that additive steering is the optimal map exactly when the target is a pure location shift, which is why the same edit that governs refusal cannot govern a behavior whose target compresses a tail.

A ceiling that can be measured before intervening. Selectivity is bounded by geometry, not by effort. In Section 3 we show that the removal an edit can buy on an axis, net of the damage it does, is capped by how well that axis separates the subpopulations it must distinguish, and in Section 5 we measure that separation on the confidence axis and find it near zero. The cap follows from a few hundred unsteered traces and costs no steered rollout, so a practitioner can know in advance that no coefficient will work, and can look for a second axis instead.

Deciding once, from a prefix. A gate that scores a finished trace and regenerates what it flags doubles the cost of serving. In Section 3 we show that a likelihood-ratio gate is optimal at every operating point, that composing it with the transport map is jointly optimal in its class, and that the same decision can be taken from a prefix of the generated trace, so the operator runs in a single pass, at $O(d)$ per token and nineteen kilobytes of state.

2 RELATED WORK

Linear steering. ActAdd (Turner et al., 2023) builds a shift vector from a single contrast prompt and adds it to the residual stream at a chosen layer. CAA (Panickssery et al., 2024) computes the same shift as a difference in means over a dataset of contrast pairs. ITI (Li et al., 2023) estimates a shift on attention heads selected by a linear probe, and RepE (Zou et al., 2023) computes reading and control vectors from prompt pairs. Directional ablation (Arditi et al., 2024) removes one direction from every activation rather than adding to it, and SAE clamping (Templeton et al., 2024) pins a sparse feature to a fixed value. All of these apply the same edit to every input, since nothing in them reads the activation before deciding how much to move it, and AxBench (Wu et al., 2025) finds the family weaker than prompting on instruction-following as a result. We take the *shape* of the edit as the object of study, and derive it instead of fixing it by hand.

Distribution matching. MiMiC (Singh et al., 2024) derives the Gaussian- W_2 affine map between two activation distributions and applies it unconditionally in the full space. Linear-AcT (Rodriguez et al., 2025a) fits a univariate transport map per neuron, controlled by a strength parameter between no transport and full transport, and LinEAS (Rodriguez et al., 2025b) learns affine maps end to end under a distributional loss. COAST (Nguyen et al., 2026) establishes pointwise optimality of a covariance-weighted projection, which in our language is transport onto a point mass, and A-LQR (Skifstad et al., 2026) establish optimality of a control input under a linear-quadratic objective. These results hold for the map alone, since none of them constrains the edit to a subspace or asks what a decision rule can add. We prove the optimal map between *distributions* under a locality constraint, and pair it with a ceiling on what any decision rule can buy.

Deciding during generation. CAST (Lee et al., 2024) compares each token’s hidden state with a condition vector at a threshold set by grid search, and applies an additive behavior vector when the comparison holds. DSAS (Ferrando et al., 2026) computes context-dependent per-token scales, and GAPS (Haider et al., 2026) applies dimension-level gates at $O(d)$ per token. Steerability can also be anticipated from a probe read after a few generated tokens (Fan et al., 2026), and YOPO (Luo et al., 2026) avoids a second clean pass by reconstructing the unsteered residual before reading its direction. Two findings in this line shape our design: reading a direction off already-steered activations degrades it (Luo et al., 2026), so our reader sits above the action, and a feature that *detects* a behavior in finished text predicts it poorly from a prefix (Kortukov et al., 2026), so we refit the head on the budget the gate will have. Our decision carries an error guarantee, since it is a likelihood-ratio test whose online form is tied to the post-hoc one at a stated rate.

3 METHOD: PROJECTION-TRANSPORT STEERING

In this section we define the object a steering operator acts on, state which operator is cost-minimal once that object is fixed (Thm. 1), bound what any decision rule on an axis can buy (Thm. 2), and show that the two can be designed separately without loss (Thm. 3).

A behavioral subspace is a small set of directions along which the behavior we want to change is legible, read off the model’s own labeled traces. Here it is spanned by two axes obtained by difference in means: a *confidence* axis, which separates confident answers from hesitant ones, and a *detection* axis, which separates the confident answers that are wrong from those that are right. The action moves the confidence coordinate and the decision reads both.

More formally, let $h \in \mathbb{R}^d$ be the residual-stream activation at a fixed decoder layer, and let $V \in \mathbb{R}^{k \times d}$ have orthonormal rows spanning that subspace, with $k=2$ in the deployed operator. We write $q = Vh \in \mathbb{R}^k$ for the projection, μ for the law of h under natural generation, μ_V for the induced law of q , and ν for a target law on $\mathbb{R}^k$. The transfer function of Table 1 is the map that takes q to its steered value, and the question of this section is which one to use.

Definition 1 (admissible steering). $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is admissible for (V, ν) if (i) $(I - V^\top V)T(h) = (I - V^\top V)h$ μ -a.s. (locality: the orthogonal complement is untouched), and (ii) $(V \circ T)_\# \mu = \nu$, i.e. the pushforward of μ equals the target law (goal: steered projections follow the target law). The cost is $C(T) = \mathbb{E}_\mu \|T(h) - h\|^2$.

We regularize by that cost because large interventions are what empirically destroy generation, as we show in Sec. 5, and because under a Fisher pullback the cost approximates the KL drift of the

next-token distribution (Prop. 3). The operator we deploy uses the Euclidean case $M_V = I_k$, so on that operator the regularizer is justified by the measurements of Sec. 5 rather than by the KL link. The first result says that once the subspace and the target law are fixed, there is nothing left to design.

Theorem 1 (steering is optimal transport of the projection). $\min_T C(T) = W_2^2(\mu_V, \nu)$ over admissible T , attained by $T^*(h) = h + V^\top(S(Vh) - Vh)$ where S is the $(\mu_V$ -a.s. unique) optimal Monge map from μ_V to ν . Any optimal T satisfies $VT(h) = S(Vh)$ almost surely, so the optimal intervention is a deterministic function of the projection alone and conditioning on $h_\perp$ cannot reduce the cost.

Proof sketch. Locality collapses the cost to the projection, and $(Vh, VT(h))$ couples (μ_V, ν) , so $C(T) \geq W_2^2$. Brenier’s map attains it. $\square$

For $k = 1$ the map $S = F_\nu^{-1} \circ F_{\mu_V}$ is the monotone quantile map, and we implement it as a 41-knot piecewise-linear lookup. For Gaussian marginals it is affine (Knott & Smith, 1984; Gelbrich, 1990), $S(q) = m_t + A(q - m_s)$ with $A = \Sigma_s^{-1/2}(\Sigma_s^{1/2}\Sigma_t\Sigma_s^{1/2})^{1/2}\Sigma_s^{-1/2}$.

Choosing the subspace. Theorem 1 holds for any orthonormal V , so k is a design parameter and not a fixed property of the behavior. Two interpretable axes make the geometry legible and are what the gate needs, but the action is free to use more: the natural larger choice is the subspace an unconditional full-space map actually moves, the mean shift $m_t - m_s$ together with the leading eigenvectors of $A - I$, at $\Theta(kd)$ per token instead of $\Theta(d^2)$. At the far end $V = I$ recovers the full-space Gaussian map that MiMiC (Singh et al., 2024) applies unconditionally, so one family runs from a single interpretable axis to a dense $\Theta(d^2)$ operator and Theorem 3 gates every point of it. We select k on the tuning split like every other hyperparameter, and on this behavior it buys less than the construction suggests (Appendix H).

Reading Theorem 1 backwards tells us when the workhorse edit is already the right one, and when it cannot be.

Corollary 1 (additive steering is forced to be global). *Additive steering $g(p) = p + \alpha$ is the optimal map iff $\nu = \mu_V * \delta_\alpha$ (pure shift). If the target compresses the spread, which is what “hedge only abnormal confidence” requires ($\sigma_t < \sigma_s$), the optimal slope is $\sigma_t/\sigma_s < 1$: the tail must move more than the bulk. A slope-1 map matched to the tail over-transportes the bulk by the same constant, the CR collapse we observe. Its selectivity between two subpopulations with equal projection laws is then identically zero, provided the probability that a trace changes state depends on h only through the projection and its displacement.*

If no function of the confidence coordinate can separate the two subpopulations, we need a second coordinate and a decision rule on it. The next result says what the best such rule is, and how much it can buy.

Theorem 2 (optimal gate and selectivity ceiling). *Let a gate fire iff a trace statistic $s > \tau$, with $\text{TPR}(\tau) = P(s > \tau \mid \text{OCW})$, $\text{FPR}(\tau) = P(s > \tau \mid \text{CR})$. (a) For any objective increasing in TPR and decreasing in FPR, the likelihood-ratio test is optimal at every operating point (Neyman–Pearson). Under equal-covariance Gaussian class conditionals it reduces to the linear discriminant $w = \Sigma^{-1}(\mu_{\text{OCW}} - \mu_{\text{CR}})$, the contrast whose error rates the statement bounds. (b) For any statistic with concavified ROC area AUC, $\max_\tau [\text{TPR} - \text{FPR}] \leq 2 \text{AUC} - 1$.*

The action can then be designed for a fixed gate event without losing displacement optimality; the gate is optimal for a different objective, and Appendix I reports what that separation does and does not buy.

Theorem 3 (optimal transport under a fixed gate event). *Among interventions that are local to V , identity off the gate event $\Gamma = \{s > \tau\}$, and achieve target law ν_Γ on Γ , the cost-minimal intervention applies Theorem 1’s map for the conditional law $(\mu_V \mid \Gamma) \rightarrow \nu_\Gamma$ on Γ ; its cost is $P(\Gamma) W_2^2(\mu_V \mid \Gamma, \nu_\Gamma)$.*

Theorem 3 treats the gate as a switch. Nothing forces that: the same cost functional, asked for the best *dose* rather than the best event, answers with a graded rule. Proposition 2 (Appendix B) gives the optimal dose as a nondecreasing function of the gate posterior $\pi(s) = P(\text{OCW} \mid s)$, affine in it wherever the expected per-state displacement is constant in s , clipped to $[0, 1]$ and zero

exactly on $\{\pi \leq \tau\}$, with τ a ratio of two measured rates: the collateral and the removal a full dose produces. The hard gate is its free-displacement limit and is strictly suboptimal at any finite price of displacement whenever π has no atom at τ . So the operating point is read off the tuning split rather than searched, and one slope is left to search in place of a threshold and an action.

Theorem 3 says nothing about *when* Γ must be decided. One way to implement the gate is to score a finished trace and then regenerate the traces it flags. Deciding while the trace is written gives another, and only the second one a server can afford. Write z_t for the mean projection over the first t generated positions and z_∞ for the completed-trace summary the post-hoc gate uses.

Theorem 4 (single-pass gating). *(a) Among all gates measurable with respect to $\sigma(z_t)$, the likelihood-ratio test on z_t is optimal at every operating point, and under equal-covariance Gaussian class conditionals it is the linear discriminant fitted on z_t — Theorem 2 applied to the statistic the gate actually has. Its ceiling is $2 \text{AUC}(z_t) - 1$, which is what a prefix budget costs. (b) If instead the gate insists on the whole-trace statistic, let the contributions form a martingale difference sequence around z_∞ with conditionally σ -sub-Gaussian increments and fire at the first t with $s_t > \tau + B_t(\delta)$, where $B_t(\delta) = 1.7 \sigma \sqrt{(\log \log 2t + 0.72 \log(5.2/\delta))/t}$ is the stitched finite-law-of-the-iterated-logarithm boundary of Howard et al. (2021). Then it fires on a trace with $s_\infty \leq \tau$ with probability at most δ , simultaneously over all t , and on a trace with margin $\Delta = s_\infty - \tau > 0$ it fires, with probability at least $1 - \delta$, by the first t with $B_t(\delta) \leq \Delta/2$, after $O(\sigma^2 \Delta^{-2} \log(1/\delta))$ positions, up to an iterated logarithm in t , positions. (c) Either way the composed operator is single-pass: the action starts at the firing position, costs $O(d)$ per token, keeps $O(1)$ extra state, and invalidates no cache entry, because positions written before the switch keep the keys they were written with.*

Proof sketch. Part (a) is Theorem 2 with z_t in place of z and assumes nothing about how z_t relates to z_∞ ; part (b) buys a time-uniform guarantee by assuming that relation, and is the confidence sequence of Howard et al. (2020; 2021), and part (c) follows from the order of the two hooks. $\square$

Remark 1 (detection and action need not share a depth). *Nothing in Theorems 1–4 forces the gate to read the layer the action writes. Placing the reader above the actor costs one position of lag and lets each sit where it is best. On gemma-2-2b-it the detection contrast peaks at layer 16, where the trace-level AUROC is 0.799 against 0.769 at layer 14, and the gap holds for the prefix heads the sequential gate uses (0.691 against 0.666 at the 32-token budget it runs at), so we let the reader sit at layer 16 while the action stays at layer 14 (Appendix G). The order also removes an interference the single-pass setting would otherwise have: reading a direction off activations the same operator has already edited degrades it (Luo et al., 2026), and here the decision is a single commitment taken before anything is edited.*

Proposition 1 (computational optimality). *Any algorithm implementing a non-degenerate intervention local to a subspace spanned by a dense direction must read all d coordinates of h , i.e. costs $\Omega(d)$ per token. PTS costs $\Theta(kd + k \log K)$, which matches that lower bound for constant k and K . The contrast that matters is not with the additive edit, which is also $\Theta(d)$, but with $\Theta(d^2)$ for a full-space affine map (MiMiC), $\Theta(d d_{\text{sae}})$ for SAE steering, and a full extra forward pass for KL-adaptive dosing. Proof in Appendix B.*

4 EXPERIMENTAL SETUP

Models, datasets, and splits. The main experiments use gemma-2-2b-it (Gemma Team, 2024), and the central comparison is repeated on gemma-2-9b-it at 2B and 9B (Sec. 6). Both models reason inside `<think>...</think>` and answer `\boxed{letter}`. Decoding is greedy, so evaluation “seeds” resample *questions*, not decoding noise. We read and modify activations at layer 14/26 (2B) and 24/42 (9B). Benchmarks: MMLU (Hendrycks et al., 2021) (fitting and primary evaluation), ARC-Challenge (Clark et al., 2018), and GSM8K (Cobbe et al., 2021) as 4-option MCQ with seeded distractors (GSM8K-MCQ, Appendix C), $n=300$ questions each, disjoint from all fitting data. We fit directions, maps, and gate parameters *once* per scale, on a 400-question MMLU extraction split ($\sim 50k$ trace tokens), and reuse them everywhere, so cross-benchmark results are transfer rather than refits. We run the main comparison on five resampled MMLU subsets and the 9B replication on three.

Behavioral states and confidence readouts. Every answer gets one of four labels — *overconfident-wrong* (OCW), *nonconfident-wrong*, *nonconfident-right*, *confident-right* (CR) — under two inde-

pendent readouts: the *letter-logit* readout M2 (softmax of the letter logits at the trace’s own answer position) and the *yes-probability* readout M4 (normalized $P(\text{YES})$ over per-option yes/no reformulations, one forward per option). The M4 prompt is given verbatim in Appendix N. On identical questions the readouts disagree sharply (ECE 0.39 against 0.10, Appendix D): M2 is strongly overconfident, M4 much better calibrated. M4 is therefore the primary readout, and agreement under the letter-logit readout is a robustness requirement; absolute counts differ between them by up to a factor of 2.6 on our headline operator. Full grids: Appendix J.

Directions and the trace-level gate. On the extraction split we fit a *confidence* axis $\hat{v}$, the steering axis, and a *detection* axis $\hat{u}$, both by diff-in-means, and orthonormalize them into $V = (\hat{v}, \hat{u})$. We keep the two contrasts disjoint by construction: the first separates the two *confident* states $\{\text{OCW}, \text{CR}\}$ from the two nonconfident ones, so $\hat{v}$ encodes confidence irrespective of correctness, and the second separates OCW from CR, so $\hat{u}$ encodes whether that confidence is warranted. A completed trace of length T is summarized by its mean projections $z = (\frac{1}{T} \sum_t \langle h_t, \hat{v} \rangle, \frac{1}{T} \sum_t \langle h_t, \hat{u} \rangle)$, and an LDA head gives the gate score $s(z) = w^\top z + b$, firing iff $s(z) > \tau$; w and b come from the equal-covariance LDA solution and τ is a quantile of the extraction-split score distribution (the median unless stated), never a quantity fitted on evaluation data. We run the gate in two regimes. *Post-hoc* (two-pass): generate unsteered, keep that output if the gate is silent, and otherwise regenerate with the intervention active during reasoning. *Sequential* (single-pass, Thm. 4): a second LDA head, fitted on the mean projection over the first t^* generated positions rather than over the finished trace, is read at layer 16 while the action runs at layer 14 (Remark 1), and the action switches on at token t^* and stays on. Fitting the head on the statistic the gate will actually have is what makes an early decision sound, and we search t^* and the threshold quantile on the tuning split. In both regimes the gate sees only the model’s own activations, with no gold labels at inference.

Steering methods. All operators act through $p_t = \langle h_t, \hat{v} \rangle$ (Sec. 3). *Additive* is $h'_t = h_t + \alpha \hat{v}$ at $\alpha = -0.75 \|h\|$, the dose that eliminates OCW under the calibrated readout. *Ablation* is $h'_t = h_t - p_t \hat{v}$, and *clamp* is $p'_t = \min(p_t, c)$ at a source quantile. *Quantile*, *Gaussian* and *Bures-Wasserstein transport* map onto a target law fitted on the extraction split. We form *conditional variants* by composing the trace gate with each action, giving CAST-style gate+additive, gated clamp, gated ablation, gated OT and LDA-gated subspace BW; in each the gate controls *exposure* and the action controls *severity*. We compare against prompting, additive CAA (Panickssery et al., 2024), ablation (Arditi et al., 2024), CAST-style (Lee et al., 2024), MiMiC (Singh et al., 2024) and Linear-AcT per-neuron transport (Rodriguez et al., 2025a), all reimplemented and run on identical records, decoding, readouts, and metrics. Every method, ours included, we give at least eight configurations in the comparison set, searched on the same held-out MMLU tuning split under one pre-declared rule: CAST gets its threshold and dose, MiMiC and Linear-AcT get layer selection, and our gate gets the depth it reads and the quantile it fires at (Appendix H).

Metrics and statistical protocol. *Accuracy* asks whether the edit hurt task performance and we report the change Δacc against the unsteered model; *ECE* asks whether the confidence numbers can be trusted. The *surgical metrics* ask whether the edit hit only what it aimed at: pairing answers by question, $\text{ocw_rm} = P(\text{left OCW} \mid \text{was OCW})$ is the share of the harmful state removed, $\text{cr_keep} = P(\text{stayed CR} \mid \text{was CR})$ the share of the healthy state preserved, and selectivity $\text{Sel} = \text{ocw_rm} - (1 - \text{cr_keep})$ combines them, +1 a perfect scalpel and 0 indiscriminate. Both are conditional on the state a question started in, so an operator can also move traces *into* the target state and we report the post-intervention population separately (Table 13). *AURC* asks whether confidence still ranks right answers above wrong ones. Single-subset results carry paired-bootstrap 95% CIs, and multi-subset results are reported as a mean with its spread over five independently resampled subsets.

5 REMOVING A BEHAVIOR WITHOUT REMOVING THE MODEL

Result: the dose that removes the behavior removes the model with it. We begin from the operator the literature ships, at the dose that meets its stated goal of eliminating overconfident-wrong answers. On the representative subset that dose leaves none of the 76 confident-right answers standing while it removes all 50 overconfident-wrong ones, at a cost of 15.0 accuracy points (Table 13). We find the damage is not confined to the benchmark: HumanEval pass@1 falls from .396 to .128, WikiText-103 perplexity doubles, and 77% of reasoning traces degenerate (Tables 6, 17). We see

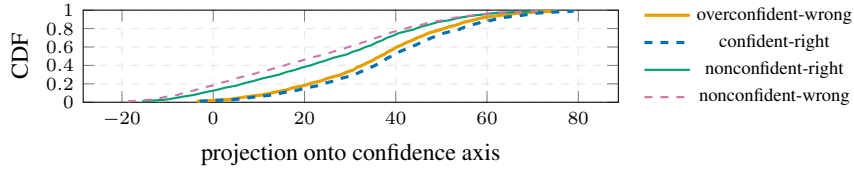


Figure 2: Token-level projection CDFs on the confidence axis by behavioral state (extraction split, $\sim 50k$ tokens). The OCW and CR curves are nearly identical (Cohen’s $d = 0.049$; $d = -0.155$ measured per trace, Table 7), while the confident and nonconfident pairs are well separated ($d \approx 0.6$). A decision rule on this coordinate is therefore close to a coin flip, which is why the gate reads a second axis.

method (M4, MMLU, seeds mean $\pm$ std)	Δ acc	ECE	AURC	ocw_rm	cr_keep	Sel
baseline	acc 0.473 $\pm$ 0.026	0.099 $\pm$ 0.019	0.404 $\pm$ 0.029	—	—	—
prompting	-0.005 $\pm$ 0.020	0.118 $\pm$ 0.018	0.392 $\pm$ 0.023	0.462 $\pm$ 0.037	0.735 $\pm$ 0.063	+0.198 $\pm$ 0.086
additive CAA Panickssery et al. (2024)	-0.087 $\pm$ 0.053	0.108 $\pm$ 0.044	0.556 $\pm$ 0.056	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	+0.000 $\pm$ 0.000
directional ablation Arditi et al. (2024)	-0.002 $\pm$ 0.038	0.074 $\pm$ 0.026	0.397 $\pm$ 0.043	0.638 $\pm$ 0.056	0.616 $\pm$ 0.057	+0.254 $\pm$ 0.109
CAST-style Lee et al. (2024)	-0.049 $\pm$ 0.039	0.099 $\pm$ 0.021	0.450 $\pm$ 0.040	0.822 $\pm$ 0.048	0.400 $\pm$ 0.037	+0.222 $\pm$ 0.055
MiMiC Singh et al. (2024)	-0.001 $\pm$ 0.041	0.116 $\pm$ 0.026	0.401 $\pm$ 0.048	0.417 $\pm$ 0.104	0.783 $\pm$ 0.054	+0.200 $\pm$ 0.107
Linear-AcT Rodriguez et al. (2025a)	+0.001 $\pm$ 0.032	0.101 $\pm$ 0.011	0.410 $\pm$ 0.027	0.279 $\pm$ 0.094	0.868 $\pm$ 0.048	+0.146 $\pm$ 0.141
clamp q_{50} (ours)	+0.001 $\pm$ 0.034	0.110 $\pm$ 0.033	0.408 $\pm$ 0.020	0.447 $\pm$ 0.045	0.698 $\pm$ 0.049	+0.145 $\pm$ 0.039
quantile transport (ours)	-0.001 $\pm$ 0.029	0.108 $\pm$ 0.017	0.408 $\pm$ 0.015	0.238 $\pm$ 0.069	0.881 $\pm$ 0.036	+0.119 $\pm$ 0.086
gated clamp (ours)	-0.001 $\pm$ 0.039	0.096 $\pm$ 0.021	0.404 $\pm$ 0.024	0.405 $\pm$ 0.061	0.820 $\pm$ 0.042	+0.224 $\pm$ 0.065
gated ablation (ours)	-0.003 $\pm$ 0.050	0.094 $\pm$ 0.030	0.395 $\pm$ 0.038	0.552 $\pm$ 0.043	0.727 $\pm$ 0.050	+0.279 $\pm$ 0.090

Table 2: **Head-to-head over five resampled MMLU subsets**, every method at its published configuration. Reading the two rows of the ablation family: the gate returns 0.111 of the confident-right answers the ungated action destroys for 0.086 less removal, at the same accuracy cost, and our gated clamp keeps 0.820 of them. Selectivity is paired on the state a question started in, so an operator can also move traces *into* the target state; the post-intervention counts are Table 13. Greedy decoding is deterministic, so seeds resample *questions*, a stricter robustness axis than sampling seeds.

the mechanism in Figure 1, and it is the one Cor. 1 names: a constant shift moves the bulk of the distribution as far as it moves the tail, so every confident state leaves the calibrated zone together.

Property: the ceiling is geometric, and it can be read before intervening. We first test the premise selective steering relies on, that the edited axis separates the states we must tell apart. We measure the two confident states as nearly coincident in Figure 2 on the confidence axis at Cohen’s $d = 0.049$, while the confident and nonconfident pairs separate at $d \approx 0.6$: the axis encodes whether the model is confident, not whether that confidence is warranted. Reading that effect size as an area under an equal-variance Gaussian ROC gives $AUC = 0.514$, and Thm. 2b turns it into a cap of 0.028 on any decision rule on this coordinate, and Cor. 1 extends it to any edit whose strength does not vary with the projection. Both come from 400 unsteered traces and cost no steered rollout. Across five resampled subsets we measure an additive selectivity of +0.000 with a standard deviation of zero,. A practitioner can therefore know in advance that no coefficient will work on this axis, and can look for a second one instead.

Result: conditioning the action on the projection returns the model. We next ask what an action that reads the projection buys. We clamp the abnormal tail, one scalar and no gate. Across five subsets we find it holds accuracy at the unsteered value and keeps 53 of the 76 confident-right answers, and quantile transport keeps 67 of them (Table 2). Both leave code synthesis, held-out perplexity and open-ended generation where they found them (Table 17). This is Cor. 1 read forwards: abnormal activations must move further than ordinary ones, and an edit of the right shape does that with no threshold to tune.

Result: the gate returns eight answers for four removals. We isolate what the gate contributes by comparing it against the strongest operator we have without one, directional ablation, the $g=0$ row of Table 1, Over five resampled subsets we find the gate gives up 4 of the 50 overconfident removals and returns 8 of the 76 correct answers, a trade of two to one at the same accuracy cost (Table 2). Figure 3 places the comparison on the removal-retention frontier: three of its four non-dominated points are ours, and the gated operators hold the high-removal end. A matched tuning budget compresses the field without reordering it: the tuned additive dose reaches the highest mean in that grid at the retention Cor. 1 says a shift must spend, 47 of 76 against our 54, and every gap in the column sits inside its spread (Appendix H). The gate buys retention, and it buys the right to leave most of the input untouched.

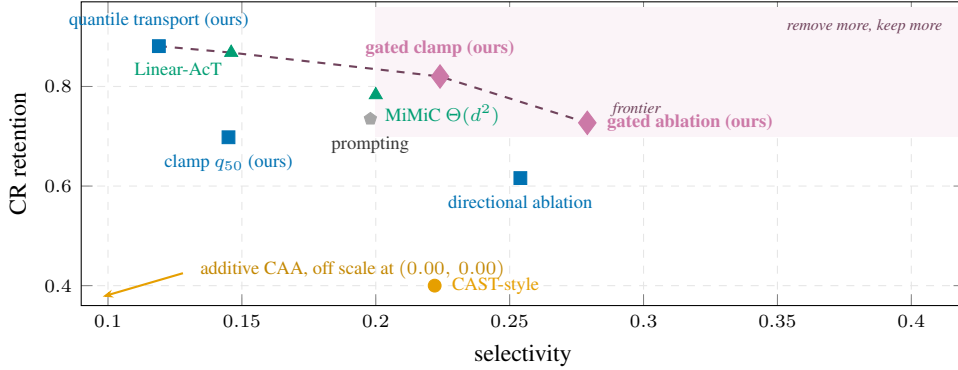


Figure 3: The removal-retention frontier. Every point is an across-subset mean of Table 2, at the published configuration, and the spreads are in that table; the dashed line is perfect retention. The additive edit (orange) buys its removal by spending every confident-correct answer, and CAST-style keeps 0.400 of them. Ungated distribution matching (green) is safe but weakly selective. The gated tail-local operators (pink) sit in the upper-right region, and directional ablation (blue) reaches comparable selectivity at 0.111 less retention.

method (M4, 9B, 3 seeds)	Δacc	ECE	ocw_rm	cr_keep	AURC	Sel
baseline	acc 0.633 $\pm$ 0.031	0.144 $\pm$ 0.021	—	—	0.197 $\pm$ 0.025	—
additive -0.75	-0.184 ± 0.033	0.167 ± 0.037	1.000 ± 0.000	0.000 ± 0.000	0.450 ± 0.023	$+0.000 \pm 0.000$
CAST-style gate+add	-0.114 ± 0.032	0.101 ± 0.011	0.831 ± 0.058	0.453 ± 0.038	0.285 ± 0.016	$+0.283 \pm 0.093$
clamp q_{50} (ours)	$+0.023 \pm 0.012$	0.143 ± 0.018	0.315 ± 0.012	0.887 ± 0.015	0.184 ± 0.025	$+0.202 \pm 0.020$
gated ablation (ours)	$+0.011 \pm 0.008$	0.127 ± 0.009	0.297 ± 0.026	0.907 ± 0.014	0.186 ± 0.023	$+0.203 \pm 0.034$

Table 3: gemma-2-9b-it, MMLU, layer 24, $n=300 \times 3$ seeds. The global shift loses 18.4 accuracy points and every confident-right answer; the CAST-style gate buys its removal with 11.4 points and 0.55 of them. Our operators gain accuracy and hold retention above 0.88.

Property: the two protections compound. Figure 1 shows where the saved answers come from. We find thirty confident-right traces fall below the gate and are returned bitwise unchanged. Of the 46 the gate does flag, 25 keep their state, because the tail-local action barely moves a typical activation. The gate alone reaches a Youden index of 0.215 on this subset while the composed operator reaches 0.304, so the action contributes on the gate’s own false positives. CAST-style has the first protection and not the second; an ungated clamp or transport has the second and not the first.

Result: the edit is invisible outside the behavior it was fitted for. An always-on edit spends most of its life on inputs we did not build it for, so we measure three tasks the directions never saw (Table 17). We find our projection-local operators at the unsteered value on every one: pass@1 between .390 and .396 against .396, perplexity between 23.1 and 23.3 against 23.4, distinct-3 between .946 and .954 against .953, and an order-consistent win rate within four points of the .5 tie value. We find the published additive dose takes pass@1 to .128 and doubles perplexity. We see the same separation in the geometry: that dose puts every MMLU token outside the 99% conditional ellipsoid, where our clamp and transport actions stay within a point of the unsteered rate (Appendix L).

6 THE OPERATOR SCALES, TRANSFERS, AND ABSTAINS

Result: scale makes the global shift costlier, not more selective. We rerun the additive, CAST-style and projection-local conditions at $4.5\times$ the parameter count (Table 3). We find the additive dose now costs 18.4 accuracy points and keeps none of the confident-right answers. We find the CAST-style gate reaches the highest selectivity in that table and pays 11.4 accuracy points and 55% of those answers for it. Our operators gain accuracy, hold retention above 0.88, and remove 30% of the target state.

Result: the operator is the exact identity where the behavior is absent. We carry the MMLU-fitted directions, maps and thresholds to two further benchmarks with no refitting (Table 4). GSM8K-MCQ carries 6 overconfident-wrong answers in 300, and we find all three hard-gated operators are the exact identity there, leaving accuracy unchanged and calibration unchanged to the third decimal. We find every unconditional operator moves accuracy on the visit, by 3.0 points for prompting and for the tuned additive dose and by 6.3 for the tuned CAST gate. We read this as

method (M4, tuned)	MMLU			ARC-Challenge			GSM8K-MCQ		
	Δ acc	ECE	Sel	Δ acc	ECE	Sel	Δ acc	ECE	Sel
baseline	—	0.076	—	—	0.113	—	—	0.258	—
prompting	−0.017	0.096	+0.097	+0.007	0.107	+0.267*	−0.030	0.195	+0.106
additive CAA (tuned)	+0.000	0.079	+0.232*	+0.003	0.084	+0.385*	−0.030	0.243	+0.000
ablation	+0.020	0.079	+0.232*	+0.020	0.110	+0.239*	−0.033	0.245	−0.151
CAST-style (tuned)	+0.007	0.076	+0.123	+0.007	0.136	+0.184*	−0.063*	0.218	+0.106
MiMiC (tuned)	+0.013	0.110	+0.070	−0.013	0.092	+0.151*	−0.013	0.233	+0.045
Linear-AcT (tuned)	+0.003	0.090	+0.182*	+0.003	0.097	+0.095	+0.023	0.269	+0.227
clamp q_{50}	+0.013	0.101	+0.124	+0.027	0.127	+0.285*	+0.027	0.289	+0.121
gated ablation (tuned, ours)	+0.027	0.094	+0.276*	+0.003	0.092	+0.290*	+0.000	0.258	+0.000
gated full-rank transport (tuned, ours)	+0.027	0.090	+0.181*	+0.003	0.105	+0.136	+0.000	0.258	+0.000
score-proportional dose (tuned, ours)	+0.043	0.091	+0.218*	−0.003	0.084	+0.342*	+0.003	0.259	+0.000
sequential gate (tuned, ours)	−0.007	0.114	+0.062	+0.020	0.109	+0.022	+0.000	0.258	+0.000

Table 4: **Cross-domain transfer with no refitting.** Unsteered accuracy is 0.480 on MMLU, 0.683 on ARC-Challenge and 0.553 on GSM8K-MCQ, so the Δ acc column is anchored on all three. Every method keeps the configuration it won on the MMLU tuning split; prompting and directional ablation carry no configuration and are run as published. * marks a bootstrap interval excluding zero, on both the Δ acc and Sel columns. GSM8K-MCQ has 6 baseline OCW and 11 CR, so its selectivity column is noisy by construction and its accuracy column is not.

the gate reading the input, not firing on a schedule: on ARC-Challenge, which carries a real over-confident tail, the same thresholds reach +0.342 at retention 0.815 and +0.290 at retention 0.901.

Result: the detector axis holds on a held-out architecture. We froze a simplified detector-axis action and evaluated it on held-out TruthfulQA panels under a pre-registered protocol (Appendix A). We find on OLMo-2 that it improves both primary metrics over a norm-matched spherical action of the same size, and passes every pre-frozen construction control, so intervention magnitude alone does not explain the effect. One of the six cells the protocol declared in advance confirmed; transfer across further models and behaviors is not established (Appendix A).

Result: one decision per trace, at $1.05\times$ the unsteered wall clock. Theorem 4 lets the same gate decide from a prefix, and we measure both sides of that on the whole 300-question workload on one idle H100 (Appendix M). We measure the sequential gate at 45.5 seconds against 43.5 unsteered. We measure scoring finished traces and regenerating the ones they flag at 85.8. A gate that commits once therefore costs a twentieth of the unsteered budget where a post-hoc gate costs another whole pass.

Property: the operator ships inside a residual-stream hook. Per token and per behavior we spend two d -dim dot products, one for the action and one for the gate, a 41-knot lookup and one FMA, $\approx 10^{-4}$ of a layer and the optimal order of Prop. 1, for a total state of ≈ 19 KB, with no extra forward pass and no KV-cache invalidation.

7 LIMITATIONS AND ASSUMPTIONS

Our work has a number of limitations. PTS relies on an axis along which the two subpopulations separate: Thm. 2 says that when none exists, no decision rule on any coordinate can be surgical and the operator degenerates to an unconditional edit. It assumes the behavior is legible in a low-dimensional projection, so that locality off the subspace costs nothing; when the behavior is carried by the complement, the constraint is the wrong one. Every deployed-operator result comes from one behavior and one model family, on five resampled subsets whose spreads differ by a factor of two, and the counts depend on the confidence readout. In future work we plan on learning the detection axis rather than reading it off contrasts, and on transport under a Fisher metric.

8 CONCLUSION

In this paper we introduce Projection-Transport Steering, a framework in which an activation edit is two separable objects: a transfer function on a behavioral subspace and a decision rule on a second axis. Once the subspace and the target law are fixed, the cost-minimal transfer function is the transport map between the projection laws, and the additive edit the literature uses is optimal

only when the target is a pure relocation. What any decision rule can buy on an axis is capped by how well that axis separates the states it must tell apart. Composing the decision with the map leaves the map optimal, and the same decision can be taken from a prefix of the generated trace.

Because the operator reads the activation before it acts, it does nothing on inputs where the behavior it targets is absent, and because it moves only the coordinates that carry that behavior, it leaves the rest of the model’s generation where it found it. The state it carries is two directions and a lookup table, so the whole operator ships inside a residual-stream hook. The most useful consequence for a practitioner needs no steered rollout at all: four hundred unsteered traces give the ceiling on what any edit along a candidate axis can achieve, and on the axis the field has been using for confidence that ceiling is 0.028.

AI USE STATEMENT

We used generative AI tools to polish the manuscript and to retrieve related work. The authors verified every resulting claim and reference and take responsibility for the final content.

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model / panel	comparison	Δ calibrated MC2 [adj. 95% CI]	Δ MC1 [adj. 95% CI]	decision
OLMo-2 / TruthfulQA	matched spherical	+ .137 [.105, +.168]	+ .105 [.025, +.177]	pass
OLMo-2 / TruthfulQA	no intervention	+ .299 [.251, +.345]	+ .287 [.177, +.384]	pass
OLMo-2 / TruthfulQA	strongest isotropic	+ .220 [.180, +.258]	+ .194 [.097, +.287]	pass
OLMo-2 / TruthfulQA	unprompted fit	+ .295 [.247, +.343]	+ .287 [.177, +.384]	pass

Table 5: Held-out OLMo-2 validation on 237 TruthfulQA questions, 50 in-domain and 187 shifted. Intervals use 10,000 paired BCa resamples and the predeclared Bonferroni family; the table shows the matched action and representative construction controls of the seven-control family, all of which pass.

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A POST-REVIEW STRESS-TEST BOUNDARY

Three terms are used below. An *observer* cell is a model/benchmark pair scored on a held-out screening split before any confirmatory test is run, and it is *eligible* when its screening effect clears a pre-declared minimum on both primary metrics; only eligible cells were opened. The *joint gate* is the pre-declared requirement that both primary metrics improve with adjusted intervals excluding zero. The *planned denominator* is the number of cells fixed before the program ran, six, which we report against whatever the outcome.

Post-review validation protocol. We froze a simplified detector-axis action after a development-only Qwen3 component analysis, and we then evaluated it on held-out TruthfulQA panels for Mistral-7B, OLMo-2-7B, and Granite-8B. Each eligible architecture used the same 237-question composition (50 in-domain, 187 shifted) and was compared with a norm-matched spherical action using 10,000 paired BCa resamples, and the three architecture comparisons used Bonferroni-adjusted two-sided intervals. We then ran seven pre-registered construction and specificity controls on OLMo-2. A separate six-cell Mistral/OLMo-2/Granite $\times$ BBQ/ETHICS observer screen kept fitting, screening, and confirmatory units disjoint, and only Mistral/ETHICS advanced to a 1,000-scenario confirmatory test, with intervals adjusted for the six planned cells. Only observer-eligible cells were opened, and the planned denominator remained six.

The confirmatory program preserved every planned cell in the denominator. Mistral/TruthfulQA improved calibrated MC2 by +.153 with adjusted 95% CI [+ .108, +.197], but its MC1 interval [−.059, +.122] crossed the predeclared joint gate; Granite/TruthfulQA stopped at observer eligibility. Five of six BBQ/ETHICS observer cells did not advance. Mistral/ETHICS did advance and improved pooled calibrated MC2 by +.107 [+ .090, +.124], but its label-1 MC1 changed by −.093 [−.124, −.071]. Thus the OLMo-2 result is a confirmed architecture transfer, while model-wide and behavior-wide generalization remain unsupported. All intervals use 10,000 paired BCa resamples with the predeclared Bonferroni family.

B FULL PROOFS

Theorem 3 treats the gate as a switch. Nothing forces that: the same cost functional, asked for the best *dose* rather than the best event, answers with a graded rule whose threshold we measure instead of tuned.

Proposition 2 (the optimal dose is affine in the gate posterior). *Let $T_\lambda(h) = h + \lambda V^\top(S(Vh) - Vh)$ interpolate between the identity and the map of Thm. 1, and let the dose λ be any $[0, 1]$ -valued function of the gate statistic s . Write $\pi(s) = P(OCW \mid s)$, $d(s) = \mathbb{E}[\|S(Vh) - Vh\|^2 \mid s]$, and let the behavioral gain of a dose be linear in it with class slopes β_1 on OCW and $-\beta_0$ on CR. Then the dose maximising gain minus displacement priced at $1/2c$ is*

$$\lambda^*(s) = \left[\frac{c(\beta_0 + \beta_1)}{d(s)} (\pi(s) - \tau) \right]_0^1, \quad \tau = \frac{\beta_0}{\beta_0 + \beta_1},$$

nondecreasing in π , zero exactly on $\{\pi \leq \tau\}$, and converging pointwise to $\mathbf{1}[\pi(s) > \tau]$ as $c/d \rightarrow \infty$. The hard gate is thus the free-displacement limit of the graded operator, and it is strictly suboptimal at any finite price of displacement whenever π has no atom at τ .

Proof sketch. The dose is s -measurable, so the objective splits as an integral over s of a strictly concave quadratic in λ ; maximise it pointwise on $[0, 1]$. $\square$

Two things follow that the switch does not give us. The operating point τ is a ratio of two measured rates, the collateral and the removal a full dose produces, so it is read off the tuning split rather than searched; and because the linear discriminant of Thm. 2a is the log-likelihood ratio, $\pi(s)$ is $\sigma(s)$ up to the prior term, which we fit on the extraction traces. One slope $\rho = c(\beta_0 + \beta_1)/d$ is left to search, in place of a threshold and an action.

Proposition 3 (metric covariance). *If cost is $(\Delta h)^\top M(\Delta h)$ with $M \succ 0$ (e.g. a Fisher/Gauss–Newton pullback, so cost approximates the induced next-token KL), the admissible problem reduces to optimal transport of μ_V to ν inside $\mathbb{R}^k$ under the Mahalanobis cost $M_V = VMV^\top \succ 0$: the optimal map is $S_M = M_V^{-1/2} \circ \nabla \varphi \circ M_V^{1/2}$ with φ convex, and the operator is again $T^*(h) = h + V^\top(S_M(Vh) - Vh)$. Euclidean PTS is $M_V = I_k$. Proof in Appendix B.*

Remark 2 (support preservation). *The quantile map never leaves the target’s empirical support. An additive map violates support for every $\alpha \neq 0$, consistent with the observed generation collapse at $|\alpha| = 1.5\|h\|$.*

Theorem 1. By locality, $T(h) - h \in \text{span}(V)$ μ -a.s., hence $\|T(h) - h\|^2 = \|VT(h) - Vh\|^2$ and $C(T) = \mathbb{E}\|VT(h) - Vh\|^2$. Let γ be the joint law of $(Vh, VT(h))$; by the goal constraint γ is a coupling of (μ_V, ν) and $C(T) = \int \|y - x\|^2 d\gamma \geq W_2^2(\mu_V, \nu)$. Since μ_V is absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^k$ with finite second moment (for $k=1$ atomlessness already suffices), Brenier’s theorem gives a unique optimal coupling induced by a map $S = \nabla \varphi$, φ convex; $T^*(h) = h + V^\top(S(Vh) - Vh)$ is admissible and attains the bound. For any optimal T , the induced coupling must equal the unique optimal one, i.e. $VT(h) = S(Vh)$ a.s. Since S depends only on Vh , the optimal intervention is $\sigma(Vh)$ -measurable: conditioning on $h_\perp$ cannot reduce the displacement cost. $\square$

Corollary 1. If $\nu = \mu_V * \delta_\alpha$ the monotone map is $x \mapsto x + \alpha$. Conversely if the optimal map is $x + \alpha$ then ν is the shift of μ_V . For Gaussians with $\sigma_t \neq \sigma_s$, after the best shift the remaining distance is $W_2^2(\mu_V * \delta_\alpha, \nu) = (\sigma_s - \sigma_t)^2 > 0$. Selectivity: let two subpopulations have the same law of p . Any transfer function g then induces the same law of displacements $\Delta = (g(p) - p)\hat{v}$ on both, so no choice of g can address one and spare the other: the operator carries no information that distinguishes them. If in addition the state-change probability depends on h only through (p, Δ) — equivalently $h_\perp$ is conditionally independent of the subpopulation given p — the two marginal change rates coincide and selectivity is zero in expectation. Without that conditional independence a residual gap can survive through $h_\perp$, but it is a property of the downstream geometry. The equality is exact only in the limit: here the two laws differ at Cohen’s $d = 0.049$, and what that residual permits does depend on the shape of g , because a map whose displacement is constant in p spends it uniformly over both subpopulations while a map whose displacement grows with p concentrates it on the tail. That is the difference between $+0.000$ for the additive dose and $+0.145$ for the median clamp in Table 2. What the corollary forbids is a selectivity that survives *exact* equality, and empirically the additive selectivity is $+0.000 \pm 0.000$ at the dose that removes the behavior (Table 2). $\square$

Theorem 2. (a) is the Neyman–Pearson lemma applied to the class-conditional laws of s ; monotone-likelihood-ratio families reduce the LR test to thresholding, and for $\mathcal{N}(\mu_{OCW}, \Sigma)$ vs $\mathcal{N}(\mu_G, \Sigma)$ the

log-LR is an affine function of the trace feature z , $\log \text{LR}(z) = w^\top z + b$ with $w = \Sigma^{-1}(\mu_B - \mu_G)$; thresholding the scalar $s(z) = w^\top z + b$ is therefore the LR test (LDA). (b) The concavified ROC contains the segment path $(0, 0) \rightarrow (\text{FPR}^*, \text{TPR}^*) \rightarrow (1, 1)$ through the Youden-optimal point; its area is $\frac{1}{2}(1 + \text{TPR}^* - \text{FPR}^*) = \frac{1+J}{2} \leq \text{AUC}$. $\square$

Theorem 3. Decompose $C(T) = P(\Gamma)\mathbb{E}[\|\Delta\|^2 \mid \Gamma] + P(\Gamma^c)\mathbb{E}[\|\Delta\|^2 \mid \Gamma^c]$. Identity off Γ zeroes the second term and is forced by the constraint; the first term with the conditional goal constraint is exactly the Theorem 1 problem for $\mu(\cdot \mid \Gamma)$. The constraints and objective separate over the partition $\{\Gamma, \Gamma^c\}$, so the piecewise optimum is global. $\square$

Proposition 2. The displacement of T_λ is $\lambda(s)^2 \|S(Vh) - Vh\|^2$, so its expectation is $\mathbb{E}_s[\lambda(s)^2 d(s)]$ because λ is s -measurable. The same measurability makes the gain $\mathbb{E}_s[\lambda(s)(\beta_1 \pi(s) - \beta_0(1 - \pi(s)))]$. The objective is therefore an integral over s of $\lambda(\beta_1 \pi - \beta_0(1 - \pi)) - \lambda^2 d/2c$, strictly concave in λ for $d(s) > 0$, and the constraint set $[0, 1]$ is separable in s , so the integral is maximised by maximising the integrand at each s . Setting the derivative to zero gives $\lambda = c((\beta_0 + \beta_1)\pi - \beta_0)/d$ and the clip enforces $[0, 1]$; factoring $\beta_0 + \beta_1$ out of the bracket gives $\pi - \tau$ with $\tau = \beta_0/(\beta_0 + \beta_1)$. Monotonicity in π and the zero set are immediate. As $c/d \rightarrow \infty$ the unclipped value diverges in sign according to $\pi - \tau$, so $\lambda^* \rightarrow \mathbf{1}[\pi > \tau]$ wherever $\pi(s) \neq \tau$, which is P -a.s. when π has no atom at τ ; at any finite c/d the objective at λ^* strictly exceeds its value at the indicator on the event $\{0 < \lambda^* < 1\}$, which then has positive probability. $\square$

Theorem 4. (a) The first t positions generate a σ -algebra $\mathcal{F}_t$; a gate measurable with respect to $\mathcal{F}_t$ is a test between the class-conditional laws of z_t , so Theorem 2(a) applies verbatim and its concavified-ROC bound gives the ceiling $2 \text{AUC}(z_t) - 1$. Nothing is assumed about how z_t relates to z_∞ : the head is fitted on z_t itself. (b) Write X_r for the gate contribution of position r and $D_r = X_r - s_\infty$. By assumption (D_r) is a martingale difference sequence with conditionally σ -sub-Gaussian increments, so $\exp(\lambda \sum_{r \leq t} D_r - \lambda^2 \sigma^2 t/2)$ is a nonnegative supermartingale for every λ Howard et al. (2020). Stitching these over geometric epochs and applying Ville’s inequality gives the finite-law-of-the-iterated-logarithm boundary and $P(\exists t : |s_t - s_\infty| > B_t(\delta)) \leq \delta$ (Howard et al. (2021), Eq. 11 at $\eta=2, s=1.4$). We measure σ rather than assume it: with $\|w\|=1$ and calibration activation norms bounded by R , Hoeffding’s lemma makes each contribution R -sub-Gaussian, and we fit the working σ to the deviation profile of the extraction traces. On $\{s_\infty \leq \tau\}$ the firing event $s_t > \tau + B_t(\delta)$ implies $s_t - s_\infty > B_t(\delta)$, which that bound excludes except on a set of measure δ ; on the complement, $s_t \geq \tau + \Delta - B_t(\delta)$, which exceeds $\tau + B_t(\delta)$ as soon as $B_t(\delta) \leq \Delta/2$, and solving $\sigma \sqrt{2 \log(\log 2t/\delta)/t} \leq \Delta/2$ gives $t = O(\sigma^2 \Delta^{-2} \log(1/\delta))$ up to an iterated logarithm in t . (c) The reader is registered on a layer at or above the actor, so within one forward pass the actor for position t observes the state accumulated over positions $< t$; the running mean needs one scalar sum and one counter per sequence, and the edit at position t changes only the keys and values written at position t , which have not yet been consumed. $\square$

Proposition 3. We argue in the latent space rather than through a global change of variables, because a general M does not preserve the duality between the measurement matrix V and the displacement subspace $\text{span}(V^\top)$. Locality writes $\Delta h = V^\top a$ with $a \in \mathbb{R}^k$, and since $VV^\top = I_k$ the goal constraint fixes $a = S(Vh) - Vh$ for whichever map S is used. Hence $\|\Delta h\|_M^2 = a^\top (VMV^\top) a = \|S(q) - q\|_{M_V}^2$ with $q = Vh$: minimising C over admissible T is exactly the Monge problem on $\mathbb{R}^k$ with cost $\|\cdot\|_{M_V}^2$. That cost is a Euclidean square in the coordinates $z = M_V^{1/2} q$ — a change of variables internal to $\mathbb{R}^k$, so the locality constraint is untouched — and Brenier’s theorem in z gives the unique optimal $\nabla \varphi$; mapping back yields $S_M = M_V^{-1/2} \circ \nabla \varphi \circ M_V^{1/2}$. For $M = I$, $M_V = I_k$ and S_M is Theorem 1’s map. $\square$

Proposition 1. *Lower bound:* let $\hat{v}_j$ be dense ($(\hat{v}_j)_i \neq 0$ for all i) and g non-constant in p_j . If an algorithm leaves coordinate i unread, an adversary perturbs h_i ; this changes $p_j = \langle h, \hat{v}_j \rangle$ and hence, on a set of positive measure, the correct output $T(h)$, while the algorithm’s output is unchanged; contradiction. Learned diff-in-means directions are dense almost surely. *Upper bound:* PTS computes $p = Vh$ (kd multiply-adds), a monotone K -knot lookup ($O(\log K)$ by binary search), and $h + V^\top(g(p) - p)$ (kd more): $\Theta(kd + k \log K)$ total. Memory $\Theta(kd + kK)$: reproducing a target marginal at quantile resolution $1/K$ requires $\Omega(K)$ parameters, and specifying the subspace requires kd . $\square$

C GSM8K MCQ CONSTRUCTION

Each GSM8K test item with integer gold g receives three deterministic distractors drawn (seeded by item index) from $\{g \pm 1, g \pm 2, g \pm 10, 2g, \lfloor g/2 \rfloor, \text{round}(1.5g), \text{round}(0.8g), g \pm \text{U}\{3..9\}\}$, deduplicated, shuffled with the gold into four options. This converts free-form arithmetic into selection while keeping the reasoning burden; we report transfer numbers on it and do not claim GSM8K free-form results.

D THE BEHAVIOR ACROSS DOMAINS AND SCALES (MEASUREMENT STUDY)

Overconfidence is model $\times$ instrument (2B). Five confidence readouts on identical MMLU questions ($n=1000 \times 3$ seeds) disagree categorically: ECE ranges from .391 [.374, .409] (letter-logit M2, the literature default) through .249/.177 (self-report/answer-distribution) to .104 [.086, .120] (per-option yes/no M4); disjoint CIs; the instruments even disagree on the answer (all five agree on 28.9% of questions). Hence our protocol requires every steering claim to hold under multiple readouts. **Domains:** on MMLU the model is overconfident (50/300 OCW), on GSM8K-MC and GPQA underconfident (6 and 27 OCW); the failure mode is domain-dependent, and the input-dependent operators inherit exactly this dependence (Table 4).

Scale replication (9B). The full pipeline was replicated on `gemma-2-9b-it` (42 layers, $d=3584$, layer 24) over the same domains. Method-dependence survives scale (MMLU ECE .126-.266, GPQA .177-.499, same instrument ranking); scale *polarizes* rather than cures (GSM8K nearly solved and calibrated, gap -0.015 ; GPQA confidence grows faster than knowledge, M2 conf .86 at acc .36); the confidence cone replicates (cosines 0.63-0.83); confidence becomes *more* linearly readable (trace-level detector AUROC to 0.83 against 0.77 at 2B), so the Thm. 2b ceiling rises with scale; and steering remains one-directional, as Cor. 1 predicts.

E WHAT EACH METHOD DOES: MEASURED MOVEMENT OF REAL TRACES

WHERE THE BENCHMARKS LIVE, AND HOW WE COMPARE WITH THE LABEL ORACLE

Figure 4 (left) plots each transfer benchmark’s baseline traces in the same plane: MMLU and ARC carry a real overconfident-wrong mass in the gated (upper-right) region, while GSM8K and GPQA have almost none. That is the geometric reason Table 4 shows activity on the former and near-identity on the latter. Figure 4 (right) compares our intervention to the *label oracle*, the conditional steering that moves exactly the overconfident-wrong points onto the calibrated centroid: our gate catches 41/50 of the oracle’s targets (recall 0.82), misses 9, and false-fires on 46/76 confident-right points (which the tail-local action then mostly spares: 25/46 keep their state). The remaining gap to the oracle is exactly the detection gap of Thm. 2b.

F SAMPLE TRACES BEFORE AND AFTER GATED ABLATION

Qualitative effects. We find gated ablation does not induce hedging: hedge markers per trace drop from 0.25 to 0.13 while traces get shorter, from 86 to 71 words, and the answer-emission rate rises from 0.91 to 0.95 (Table 6).

On the 223/300 gated questions, gated ablation does not induce hedging: hedge markers *drop* from 0.25 to 0.13 per trace and length from 86 to 71 words, 29 of the 50 OCW leave that state and 12 come back correct. Typical outcomes: an associative wrong argument replaced by an explicit derivation, better-calibrated reasoning with the same letter, and a flagged CR rephrased but preserved (strong additive or CAST doses instead yield repetition loops). The excerpts below are truncated, and complete rollouts ship with the released artifact.

Across the 223/300 gated questions, the intervention does *not* inject hedging vocabulary — hedge-marker frequency *drops* (0.25 $\rightarrow$ 0.13 per trace) — and traces get shorter (86 $\rightarrow$ 71 words). What changes is the mode of reasoning: assertive shortcut arguments are replaced by definitional, evidence-first ones. Of the 50 overconfident-wrong traces, 29 leave that state and 12 come back *correct* under M4, while 55 of the 76 confident-right traces keep their state. Ablating the confidence

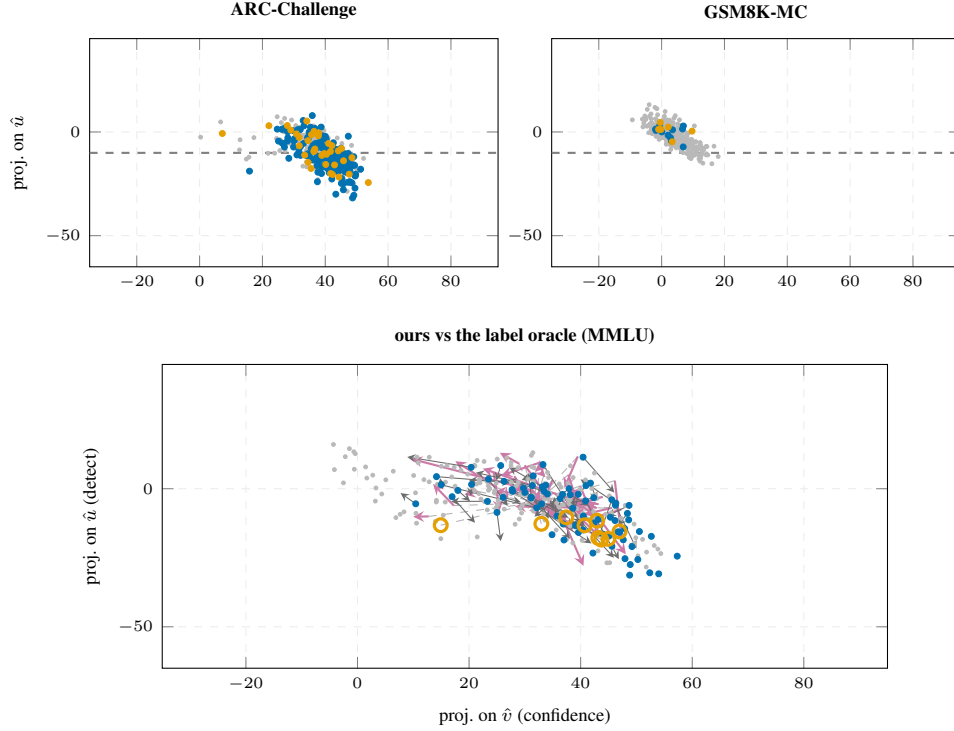


Figure 4: Top: transfer benchmarks in the behavioral plane (gate threshold dashed): ARC has an overconfident-wrong tail above the gate, GSM8K almost none, and transfer activity follows the geometry. Bottom: gray dashed arrows = the label oracle (move every overconfident-wrong point to the calibrated centroid); pink = our measured moves on the same points (41/50 caught); hollow orange circles = the 9 misses; dark-gray thin arrows = false fires on confident-right (mostly state-preserving). The oracle–ours difference is the detection gap that bounds selectivity (Thm. 2b).

condition (MMLU, M2 traces)	words	closes	boxed	distinct-3	degen.%	hedge	certainty
unsteered	86	0.98	0.91	0.91	2.7	0.25	0.12
additive -0.75 Panickssery et al. (2024)	186	0.28	0.20	0.31	76.7	0.08	0.29
clamp q_{50} (ours)	76	0.99	0.94	0.91	2.7	0.18	0.10
gated ablation (ours)	71	0.98	0.95	0.90	3.7	0.13	0.15
sequential gate (ours)	79	0.98	0.93	0.91	2.3	0.24	0.11

Table 6: What each operator does to the reasoning trace. “Closes” is the fraction of traces that emit `</think>` and “boxed” the fraction that emit an answer; degen.% counts traces with distinct-3 below 0.5; hedge and certainty are marker counts per trace from the fixed lexicons of Appendix N.

projection during the `think` phase appears to remove a “commit early” pressure rather than adding doubt. Three representative pairs (traces truncated):

Overconfident-wrong $\rightarrow$ corrected (mmlu-4592; gold C, M4 state overconfident wrong $\rightarrow$ non-confident wrong). The baseline trace jumps from a vague association to an answer; the steered trace states the definition and derives the (gold) answer.

Question: If the price of firm A’s cell phone service rises by 5 percent and the quantity demanded for firm B’s cell phone service increases by 10 percent, we can say that Options: A. demand for firm B is price elastic. B. supply

before: `<think>`The question asks about the relationship between the price of firm A’s service and the quantity demanded for firm B’s service. We know that the price of firm A’s service increased and the quantity demanded for firm B’s service increased. This suggests that the demand for firm B’s service is likely to be price elastic. `</think>` `\boxed{A}` `\rightarrow A`

after: `<think>`The cross-price elasticity of demand measures the percentage change in the quantity demanded of one good in response to a percentage change in the price of another good. Since the price of firm A’s service increased and the quantity demanded for firm B increased, the cross-price elasticity is positive. We can conclude that the two goods are substitutes. `</think>` `\boxed{C}` `\rightarrow C`

Overconfident-wrong → **calibrated-right** (mmlu-12923; gold C, M4 state overconfident wrong → nonconfident right). The baseline reopens `<think>` three times and re-states the same elimination in each block, so it runs out of budget before reaching D; the steered trace weighs all four options once and lands on the same letter with calibrated confidence.

Question: The best conclusion that can be drawn about the four-day (compressed) workweek is that it has: Options: A. long-lasting positive effects on attitudes and productivity. B. long-lasting negative effects on attitudes and productivity. We need to consider the potential effects on attitudes and productivity. There's no mention of specific evidence or studies, so we need to consider the general impact of a compressed workweek. `</think>`
before: `<think>`The question asks for the best conclusion about the four-day workweek. We need to consider the potential effects on attitudes and productivity. There's no mention of specific evidence or studies, so we need to consider the general impact of a compressed workweek. `</think>`
`<think>`Option A is too strong. We don't have enough information to say it has long-lasting positive effects on attitudes and productivity. `</think>`
`<think>`Option B is too strong. We don't have enough information to say it has long-lasting negative effects on attitudes and productivity. `</think>`
after: `<think>`The question asks for the best conclusion about the four-day workweek. We need to consider the potential effects on attitudes and productivity. Option A is too strong, as we don't have enough information to guarantee long-lasting positive effects. Option B is too strong, as we don't have enough information to guarantee long-lasting negative effects. Option C is a possibility, but we don't know if it's the best conclusion. Option D is a possibility, but we don't know if it's the best conclusion. `</think>`
`\boxed{D}` → D

Confident-right, gated but spared (mmlu-5125; gold B, M4 state confident right → confident right). A false-flagged trace whose state survives the intervention. The state label is the M4 readout, which puts its mass on the gold option B; the displayed rollout is the M2 trace, whose greedy letter is A, so instrument and greedy letter disagree here (Appendix D). Under steering the two agree: the trace states Pavlov's discovery and emits B.

Question: Which one of the following best states Ivan Pavlov's accidental discovery? Options: A. While experimenting with rats, Pavlov discovered that if he repeatedly placed pellets of food in one side of a "T-maze," the rats would learn to go to that side. B. While experimenting with rats, Pavlov discovered that if he repeatedly placed pellets of food in one side of a "T-maze," the rats would learn to go to the other side. C. While experimenting with rats, Pavlov discovered that if he repeatedly placed pellets of food in one side of a "T-maze," the rats would learn to go to the middle. D. While experimenting with rats, Pavlov discovered that if he repeatedly placed pellets of food in one side of a "T-maze," the rats would learn to go to the end.
before: `<think>`The question asks for Pavlov's accidental discovery. Let's look at the options and see which one describes an accidental discovery. Option A describes a learned response to a stimulus, which is not an accidental discovery. Option C describes a cat learning to press a lever, which is a learned behavior. Option D describes a pigeon learning to associate a key with food. Option B describes a rat learning to run to a specific side of a maze. This is a classic example of classical conditioning. Therefore, → A
after: `<think>`The question asks for the accidental discovery of Pavlov. Option B is the best choice because it describes Pavlov's accidental discovery of the dog's salivation in response to the sight and sound of the attendant who brought them food. `</think>` `\boxed{B}` → B

ONE QUESTION, FOUR METHODS

The most instructive comparison holds the question fixed (mmlu-4592, gold C; baseline answers A, overconfident-wrong). Our gated ablation produces the clean definitional trace shown above and lands on C. The additive-action methods do something qualitatively different — their traces *degenerate into verbatim repetition loops*, the signature of support violation (Remark 2): the constant shift pushes activations off the data manifold and generation cycles. Trace length balloons from 351 to ~1500 characters of looped text; this is not deliberation but decoding pathology, invisible in aggregate confidence numbers yet obvious in the rollouts. MiMiC (distribution-matched, unconditional) stays fluent but keeps the wrong answer.

additive −0.75 Panickssery et al. (2024) (→ A):

`<think>`The cross-price elasticity of the cell phone service is equal to the price of the cell phone service of the firm A and the quantity of the cell phone service of the firm B. The cross-price elasticity of the cell phone service is equal to the price of the cell phone service of the firm A and the quantity of the cell phone service of

CAST-style gate+add Lee et al. (2024) (→ A):

`<think>`The cross-price elasticity of demand for firm B is $(10/5) = 2$. The cross-price elasticity of demand for firm A is $(10/5) = 2$. The cross-price elasticity of demand for firm A is $(10/5) = 2$. The cross-price elasticity of demand for firm B is $(10/5) = 2$. The cross-price elasticity of demand for firm A is $(10/5) = 2$. The cross-price el

layer	Cohen’s d conf.	Cohen’s d OCW/CR	gate AUROC	ceiling 2AUC-1
4	+0.525	-0.079	0.783	+0.566
7	+0.535	-0.093	0.756	+0.511
10	+0.512	-0.144	0.741	+0.482
12	+0.516	-0.150	0.742	+0.485
14	+0.589	-0.155	0.769	+0.538
16 *	+0.631	-0.141	0.799	+0.599
18	+0.616	-0.116	0.779	+0.558
22	+0.675	-0.110	0.791	+0.582
25	+0.700	-0.111	0.649	+0.298

Table 7: Layer sweep on the extraction split (400 traces, both axes and the LDA gate refit at each depth). * marks the depth the sequential gate reads. The OCW/CR column is measured on the axis refit at that depth and is per trace rather than per token, so it is not the $d = 0.049$ of Fig. 2, which is a token-level statistic over $\sim 50k$ positions. The two estimates differ in sign and the trace-level one is the larger in magnitude; at $|d| = 0.155$ the ceiling of Thm. 2b is still only 0.087, so the argument the small effect size supports does not turn on which estimate is used.

MiMiC Singh et al. (2024) ($\rightarrow$ A):

<think>The question asks about the relationship between the price of firm A’s service and the quantity demanded for firm B’s service. We know that the price of firm A’s service increased and the quantity demanded for firm B’s service increased. This suggests that the demand for firm B’s service is

gated ablation (ours) ($\rightarrow$ C):

<think>The cross-price elasticity of demand measures the percentage change in the quantity demanded of one good in response to a percentage change in the price of another good. Since the price of firm A’s service increased and the quantity demanded for firm B increased, the cross-price elasticity is positive. We can conclude that the tw

G WHERE TO READ AND WHERE TO ACT: THE LAYER SWEEP

Where the gate reads. We never required the gate to read the layer the action writes (Remark 1), and refitting it at nine depths shows the contrast is not monotone: it peaks at layer 16 at AUROC 0.799 against 0.769 at layer 14, and layer 16 leads layer 14 at all six prefix budgets. The sequential gate takes that split; the post-hoc gate the tuning split selected reads at layer 14 (Appendix G). We designed the gate to be permissive — it fires on 223 of 300 MMLU questions — and what keeps the edit local is that the action does almost nothing to the traces it wrongly flags. Five alternative actions inside the same gate span +0.042 to +0.231 in selectivity, so gate and action have to be chosen together.

Where the gate reads, and where the action writes. The framework never required the gate to read the layer the action writes (Remark 1), so we ask where the contrast is most readable. We refit the gate at nine depths (Appendix G) and find the separation is not monotone: it peaks at layer 16 at AUROC 0.799 against 0.769 at layer 14, and falls by layer 25 to 0.649, where the residual stream has already committed to an answer. Reading at layer 16 while acting at layer 14 buys that gap for one position of lag, and a larger one for the prefix heads the sequential gate uses. The gate is permissive by design: at its selected threshold it fires on 223 of 300 MMLU questions, and what keeps the edit local is that the action does almost nothing to the traces it wrongly flags. Five alternative actions inside the same gate span +0.042 to +0.231 in selectivity (Appendix I), so the action is not interchangeable either, and the two choices have to be made together.

Table 7 refits both axes and the gate at nine depths of `gemma-2-2b-it` on the same 400 extraction traces. Two quantities move in opposite directions. Token-level confidence separation broadly grows with depth (Cohen’s d from 0.53 to 0.70), because late layers carry the answer itself. Detector quality, the quantity Theorem 2b turns into a selectivity ceiling, peaks in the middle and collapses in the last layers, where the residual stream has already committed. Layer 14 is inside the plateau and layer 16 is its maximum; the gap between them, 2 AUROC - 1 of 0.538 against 0.599, is why the sequential gate reads two layers above the layer it edits (Remark 1).

gate AUROC (OCW vs CR)	$t=8$	16	32	64	128	256
layer 14	0.708	0.712	0.666	0.707	0.747	0.766
layer 16	0.736	0.740	0.691	0.738	0.783	0.795

Table 8: Detector quality as a function of the generated-token budget the gate is allowed to see, with the head refitted for each budget. The whole-trace values are 0.769 at layer 14 and 0.799 at layer 16.

WHAT A PREFIX BUDGET COSTS THE GATE

A head fitted on completed traces is a detector, and a detector is not automatically a predictor: features that recognise a behavior in finished text forecast it poorly (Kortukov et al., 2026) from a prefix Kortukov et al. (2026). We therefore do not reuse the whole-trace head online. For each budget t we refit the LDA on the mean projection over the first t generated positions and measure its AUROC on the same contrast (Table 8). Deciding early costs less than a prefix budget might suggest: at layer 16 a head that sees eight generated tokens reaches 0.736 against 0.799 for the whole trace, and by 128 tokens the gap is 0.016. The dip at $t=32$ is sampling noise in a two-feature fit on 400 traces and does not survive at either neighbour. Layer 16 dominates layer 14 at every budget, which is why the sequential gate reads there.

H EQUAL-BUDGET TUNING

Under this budget the tuned additive dose attains the highest mean selectivity $+0.256 \pm 0.133$ and the untuned directional ablation the second $+0.254 \pm 0.109$, with our graded dose third $+0.231 \pm 0.120$, the tuned CAST gate fourth $+0.227 \pm 0.082$ and our gated ablation fifth $+0.206 \pm 0.059$. The gaps are inside the spreads, which differ by more than a factor of two. Deciding from a 32-token prefix rather than a finished trace takes our selectivity from $+0.206 \pm 0.059$ to $+0.079 \pm 0.050$ at retention 0.897.

What equal tuning changes, and what it does not. The first question to ask of a head-to-head is whether the baselines were tuned as hard as the method, so we gave every operator with a searchable configuration at least as many as our own eight, on the same held-out split of $n=1200$ questions, under one pre-declared rule (Appendix H, Table 11). We find tuning compresses the field. The additive dose falls to a third of the published one and reaches $+0.256 \pm 0.133$, directional ablation sits second at $+0.254 \pm 0.109$ without a configuration to search, the tuned CAST gate reaches $+0.227 \pm 0.082$, our score-proportional dose $+0.231 \pm 0.120$ and our gated ablation $+0.206 \pm 0.059$. We do not read the head of that column as a result, because the gaps are inside the spreads. We observe that the spreads themselves differ by more than a factor of two, ± 0.059 against ± 0.133 , which five subsets are too few to test. What separates the families is the mechanism Cor. 1 names — the tuned dose keeps 0.621 of the confident-correct answers — and what happens when the domain changes.

The subspace dimension is not the lever. To find out where in the family the strength lies, we tuned it end to end. Theorem 3 says our gate composes with any transport, so a full-space map is not a rival family but the $k=d$ member of ours, and we tuned the family end to end under one budget. We measure selectivity peaking at $k=2$, $+0.186$, and flat above it, $+0.148$ at $k=8$ and again at $k=32$, so the two interpretable axes already carry this behavior and the extra 2302 coordinates buy nothing. The $k=d$ member says the same thing from the other end. We measure the gate composed with MiMiC’s own map at $+0.286$ on a 400-question split and $+0.155$ on the 1200-question one, the largest movement of any configuration we searched (Table 10), the signature of a configuration selected on noise once that noise is removed. We keep $k=2$ because the data says the behavior lives there, not because the operator is cheaper there. The other members of the family, a token-local action and a posterior-weighted dose, we tune and carry the same way in Appendix I.

Every method in Table 2 receives at least as many configurations as our own eight, on the same held-out MMLU tuning split of $n=1200$ questions, disjoint from the extraction split and from all five evaluation subsets, and we select among them by one pre-declared rule, the highest selectivity among configurations whose accuracy change is at least -0.01 . The additive baseline searches its dose, CAST searches threshold $\times$ dose, MiMiC searches layer $\times$ shrinkage, Linear-Act searches layer $\times \lambda$, our conditional ablation searches the depth the gate reads $\times$ the quantile it fires at $\times$ the local action, our transport variant searches the subspace dimension $k \times$ the gate quantile, and the sequential gate searches its prefix budget $t^* \times$ the gate quantile. We report each winner in Table 9,

method	budget	selected configuration	Δacc	ocw_rm	cr_keep	Sel
additive CAA Panickssery et al. (2024)	8	alpha-0.25	+0.016	0.627	0.644	+0.271
CAST-style Lee et al. (2024)	8	tau_cr_q70_alpha-0.375	-0.005	0.428	0.756	+0.184
MiMiC Singh et al. (2024)	9	L18_reg0.1	+0.030	0.468	0.795	+0.263
Linear-AcT Rodriguez et al. (2025a)	9	L14_lam1.0	+0.026	0.318	0.865	+0.184
gated ablation (tuned, ours)	8	g114_q30_ablate	+0.001	0.527	0.718	+0.245
gated k -dim transport (ours)	8	k2_q30	+0.005	0.279	0.907	+0.186
score-proportional dose (ours)	8	g116_q10_kappa1.0	+0.010	0.607	0.660	+0.267
sequential gate (ours, single pass)	8	q50_at32	+0.014	0.219	0.917	+0.136
NP-gate $\times$ MiMiC (Thm. 3)	4	L14_q30_mimic	+0.022	0.318	0.837	+0.155

Table 9: Equal-budget hyperparameter search on the tuning split.

method	$n=400$ split		$n=1200$ split		
	winner	Sel	winner	Sel	Δ
additive CAA Panickssery et al. (2024)	alpha-0.25	+0.196	alpha-0.25	+0.271	+0.075
CAST-style Lee et al. (2024)	tau_cr_q70_alpha-0.75	+0.182	tau_cr_q70_alpha-0.375	+0.184	+0.003
MiMiC Singh et al. (2024)	L18_reg0.01	+0.288	L18_reg0.1	+0.263	-0.025
Linear-AcT Rodriguez et al. (2025a)	L10_lam0.25	+0.099	L14_lam1.0	+0.184	+0.085
gated ablation (tuned, ours)	g114_q30_ablate	+0.232	g114_q30_ablate	+0.245	+0.013
gated k -dim transport (ours)	k32_q30	+0.165	k2_q30	+0.186	+0.021
score-proportional dose (ours)	g116_q10_kappa0.5	+0.265	g116_q10_kappa1.0	+0.267	+0.003
sequential gate (ours, single pass)	q30_at16	+0.120	q50_at32	+0.136	+0.015
NP-gate $\times$ MiMiC (Thm. 3)	L18_q30_mimic	+0.286	L14_q30_mimic	+0.155	-0.131

Table 10: The same search on a tuning split three times larger. Seven of the nine winners change; only additive CAA (alpha-0.25) and our gated ablation (g114_q30_ablate) select the same configuration on both splits, and even for those two the measured selectivity moves by up to 0.075. A 400-question split does not identify a configuration reliably. Prompting and directional ablation carry no searchable configuration in our implementation and are run as published. The paper uses the $n=1200$ winners throughout.

method (M4, MMLU, seeds mean $\pm$ std)	Δacc	ECE	AURC	cr_keep	Sel
baseline	acc 0.473 $\pm$ 0.026	0.099 $\pm$ 0.019	0.404 $\pm$ 0.029	—	—
prompting	-0.005 $\pm$ 0.020	0.118 $\pm$ 0.018	0.392 $\pm$ 0.023	0.735 $\pm$ 0.063	+0.198 $\pm$ 0.086
additive CAA Panickssery et al. (2024)	-0.004 $\pm$ 0.041	0.080 $\pm$ 0.017	0.400 $\pm$ 0.038	0.621 $\pm$ 0.067	+0.256 $\pm$ 0.133
directional ablation Arditì et al. (2024) (as published)	-0.002 $\pm$ 0.038	0.074 $\pm$ 0.026	0.397 $\pm$ 0.043	0.616 $\pm$ 0.057	+0.254 $\pm$ 0.109
CAST-style Lee et al. (2024)	-0.007 $\pm$ 0.032	0.072 $\pm$ 0.012	0.396 $\pm$ 0.017	0.795 $\pm$ 0.031	+0.227 $\pm$ 0.082
MiMiC Singh et al. (2024)	-0.015 $\pm$ 0.045	0.113 $\pm$ 0.010	0.408 $\pm$ 0.040	0.763 $\pm$ 0.031	+0.132 $\pm$ 0.103
Linear-AcT Rodriguez et al. (2025a)	+0.001 $\pm$ 0.032	0.101 $\pm$ 0.011	0.410 $\pm$ 0.027	0.868 $\pm$ 0.048	+0.146 $\pm$ 0.141
gated ablation (tuned, ours)	-0.015 $\pm$ 0.036	0.089 $\pm$ 0.013	0.413 $\pm$ 0.037	0.713 $\pm$ 0.050	+0.206 $\pm$ 0.059
gated full-rank transport (ours)	+0.003 $\pm$ 0.021	0.108 $\pm$ 0.022	0.394 $\pm$ 0.044	0.854 $\pm$ 0.038	+0.146 $\pm$ 0.042
score-proportional dose (ours)	-0.005 $\pm$ 0.054	0.089 $\pm$ 0.010	0.406 $\pm$ 0.050	0.650 $\pm$ 0.056	+0.231 $\pm$ 0.120
sequential gate (ours, single pass)	+0.009 $\pm$ 0.015	0.111 $\pm$ 0.018	0.398 $\pm$ 0.024	0.897 $\pm$ 0.017	+0.079 $\pm$ 0.050

Table 11: **Matched-budget grid.** The same five evaluation subsets with every method carrying the configuration it won under the budget of this appendix; directional ablation carries no searchable configuration in our implementation and is shown at its published setting. The tuned additive dose reaches the highest mean, $+0.256 \pm 0.133$, by retaining 0.621 of CR; our score-proportional dose reaches $+0.231 \pm 0.120$ and our gated ablation $+0.206 \pm 0.059$ at retention 0.713. The tuned CAST gate reaches $+0.227 \pm 0.082$ by removing 0.432 of the OCW it fires on. Those are the two ways Cor. 1 says a shift can trade.

and those configurations, unchanged, produce the evaluation-seed numbers in the main text. The split has to be that large. On a 400-question split seven of the nine winners are different, and among the two configurations that do hold fixed the measured selectivity still moves by up to 0.075 (Table 10), so the smaller split does not identify a configuration, it samples one. Every search also runs at one batch size, because batched greedy decoding is not invariant to it and the same MiMiC configuration measures $+0.288$ at 64 and $+0.235$ at 32. Each method’s search is paired against a baseline generated in the same process, which is what config selection needs; the searches ran on different GPUs and their baselines differ by up to three OCW out of 79, so we do not compare selectivities *across* methods on this split. The equal-budget cross-method comparison is Table 11, where every winner is carried unchanged to five evaluation subsets and paired against one shared set of baseline rollouts per seed. We also report one configuration that is not part of any selection, the NP gate composed with the full-space affine map, because Theorem 3 predicts the gate composes with any transport and that is the cheapest way to test the prediction on the strongest unconditional action.

member (M4, MMLU, seeds mean $\pm$ std)	Δ acc	ECE	cr_keep	Sel
baseline	acc 0.473 $\pm$ 0.026	0.099 $\pm$ 0.019	—	—
gated ablation (tuned), $k=2$	-0.015 $\pm$ 0.036	0.089 $\pm$ 0.013	0.713 $\pm$ 0.050	+0.206 $\pm$ 0.059
gated k -dim transport	-0.010 $\pm$ 0.012	0.100 $\pm$ 0.011	0.882 $\pm$ 0.053	+0.065 $\pm$ 0.095
gated token-local clamp	-0.008 $\pm$ 0.053	0.095 $\pm$ 0.023	0.721 $\pm$ 0.050	+0.158 $\pm$ 0.085
score-proportional dose	-0.005 $\pm$ 0.054	0.089 $\pm$ 0.010	0.650 $\pm$ 0.056	+0.231 $\pm$ 0.120
posterior-weighted dose	-0.003 $\pm$ 0.006	0.093 $\pm$ 0.017	0.969 $\pm$ 0.027	+0.042 $\pm$ 0.050
gated full-rank transport	+0.003 $\pm$ 0.021	0.108 $\pm$ 0.022	0.854 $\pm$ 0.038	+0.146 $\pm$ 0.042

Table 12: Members of the projection-transport family under the equal-budget protocol, each at the configuration it won on the $n=1200$ tuning split.

I DESIGN ABLATIONS OF THE OPERATOR FAMILY

Spending the edit where the evidence is. The same cost functional, asked for the best dose rather than the best event, gives the graded operator of Prop. 2, tuned under the same budget. We find the selected configuration widens exposure instead of narrowing it, firing at the 10th percentile and scaling each edit by its margin, so grading the dose is what makes a permissive threshold affordable. On ARC-Challenge, off the split it was tuned on, it is the best member we have at +0.342 with retention 0.815 against +0.290 for the hard switch.

We tuned every member of the family Theorem 1 defines under the same budget and carried each unchanged to the five evaluation subsets. The point of the table is which design choices survive that carry. Raising the subspace dimension does not: the tuned k falls to +0.065 on the subsets from +0.186 on the split. Making the action token-local rather than trace-local does not either, +0.158 from +0.257, and neither does weighting the dose by a posterior. Grading the dose by the gate margin does survive, +0.231 from +0.267, and so does the plain $k=2$ gated ablation, +0.206 from +0.245. The two lose 0.036 and 0.039, which we read as indistinguishable on this axis. One member is worth reporting on its own terms. Theorem 3 gives the cost-minimal gated intervention as the transport map of the law *conditioned* on the gate event, and every gated transport above uses the marginal law instead. We fitted the conditional moments on the extraction split, at each of the four gate quantiles, and compared the two maps under one budget. The conditional map is not better: +0.155 against +0.183 at the 20th percentile, +0.153 against +0.125 at the 30th, and no systematic ordering across the four. We read that as Theorem 3 delivering what it claims and nothing else. It minimises displacement subject to reaching the target law, which is not the same objective as maximising selectivity, and selectivity here is bound by the detector rather than by the map (Thm. 2b). Five attempts to buy more removal by changing the action — a larger subspace, a full-rank map, a graded dose, a token-local clamp and the conditional map — each bought it at the same rate in retention, which is what Cor. 1 predicts for a subpopulation pair separated by $d = 0.049$.

We report the members that did not carry because each one is an insight about the selection problem rather than a catalogue entry. Together they say that the more exposure or dose a configuration buys on a tuning split, the less of it survives a change of questions, and that the design choice which survives best is the one the theory singles out: a two-dimensional edit applied behind a gate.

J FULL RESULT TABLES

The instruments disagree on the middle of the table, not on the head. On the representative subset we measure gated ablation at +0.304 under the calibrated yes-probability readout and +0.119 under the letter-logit readout, whose baseline OCW population is 124 rather than 50. Four conditions change the sign of their selectivity between the two grids and several middle-ranked methods swap places, but ours is the most selective of the published-configuration set under both (Appendix J).

Robustness across instruments. Absolute counts depend on the readout. On the representative subset gated ablation reaches selectivity +0.304 under the calibrated yes-probability readout M4 and +0.119 under the letter-logit readout M2, a factor of 2.6. The two grids do not agree everywhere: four conditions change the sign of their selectivity and several middle-ranked methods swap places. Our operator is the most selective of the published-configuration set under both, and its accuracy column is within noise of zero under both.

condition (M4 readout)	Δ acc	95% CI	ECE	OCW after	cr_keep	Sel [95% CI]
baseline	—	—	0.076	50	—	—
additive -0.75 Panickssery et al. (2024)	-0.150	$[-0.21, -0.08]$	0.052	0	0.000	$+0.000$ $[+0.00, +0.00]$
ablation Arditi et al. (2024)	$+0.020$	$[-0.04, +0.08]$	0.079	30	0.592	$+0.232$ $[+0.07, +0.40]$
clamp q_{50}	$+0.013$	$[-0.04, +0.07]$	0.101	48	0.724	$+0.124$ $[-0.05, +0.30]$
quantile transport	-0.007	$[-0.04, +0.03]$	0.096	54	0.829	$+0.029$ $[-0.12, +0.18]$
gated clamp q_{50}	$+0.017$	$[-0.03, +0.07]$	0.081	47	0.855	$+0.195$ $[+0.04, +0.35]$
gated clamp q_{70}	-0.003	$[-0.04, +0.04]$	0.045	39	0.895	$+0.175$ $[+0.04, +0.32]$
gated ablation	$+0.033$	$[-0.02, +0.09]$	0.101	35	0.724	$+0.304$ $[+0.13, +0.47]$
sequential-gate ablation	$+0.003$	$[-0.03, +0.04]$	0.098	49	0.934	$+0.074$ $[-0.03, +0.19]$

Table 13: MMLU, $n=300$, layer 14, M4 readout. The OCW column is the post-intervention population, not the number removed: because selectivity is paired on the state a question started in, an operator can also move traces *into* the target state, and quantile transport ends with 54 against a baseline of 50 while scoring $+0.029$. Gated ablation removes 29 of the 50 it starts with and ends at 35.

The full condition grids follow. Columns: condition, readout, Δ accuracy with paired-bootstrap 95% CI, ECE, count of OCW after steering, CR-retention, and selectivity with its interval.

Table 14: MMLU: all conditions, both readouts. sweep_ocwcr-crq50_ablate is the gated ablation of Table 2, m4_conf_otq_nonocw its quantile transport and m4_conf_gate-ocw-vs-cr-crq50_clamp-q50 its gated clamp; Appendix N maps every run identifier to the name used in the text.

condition	r/o	Δ acc [CI]	ECE	OCW	cr_keep	Sel [CI]
baseline	m4	—	0.076	50	—	—
Linear-AcT $\lambda=0.5$	m4	-0.007 $[-0.053, +0.037]$	0.084	50	0.83	$+0.109$ $[-0.04, +0.25]$
Linear-AcT $\lambda=1.0$	m4	$+0.003$ $[-0.037, +0.043]$	0.090	48	0.88	$+0.182$ $[+0.03, +0.33]$
CAST-style gate+add	m4	-0.067 $[-0.130, -0.003]$	0.079	9	0.39	$+0.215$ $[+0.05, +0.37]$
ablation	m4	$+0.020$ $[-0.040, +0.080]$	0.079	30	0.59	$+0.232$ $[+0.07, +0.40]$
additive -0.75	m4	-0.150 $[-0.213, -0.080]$	0.052	0	0.00	$+0.000$ $[+0.00, +0.00]$
clamp q_{30}	m4	$+0.027$ $[-0.030, +0.083]$	0.102	40	0.68	$+0.184$ $[+0.01, +0.36]$
clamp q_{50}	m4	$+0.013$ $[-0.043, +0.070]$	0.101	48	0.72	$+0.124$ $[-0.05, +0.30]$
clamp q_{70}	m4	$+0.023$ $[-0.030, +0.077]$	0.136	48	0.72	$+0.144$ $[-0.03, +0.31]$
gated clamp q_{50}	m4	$+0.017$ $[-0.033, +0.067]$	0.081	47	0.86	$+0.195$ $[+0.04, +0.35]$
gated OT q_{50}	m4	-0.023 $[-0.067, +0.020]$	0.095	50	0.84	$+0.042$ $[-0.10, +0.18]$
gated clamp q_{70}	m4	-0.003 $[-0.043, +0.040]$	0.045	39	0.89	$+0.175$ $[+0.04, +0.32]$
gated OT q_{70}	m4	-0.010 $[-0.037, +0.017]$	0.083	53	0.92	$+0.001$ $[-0.09, +0.10]$
Gaussian-OT cal.	m4	$+0.027$ $[-0.010, +0.067]$	0.119	48	0.88	$+0.062$ $[-0.06, +0.19]$
quantile-OT cal.	m4	$+0.007$ $[-0.030, +0.047]$	0.131	55	0.84	$+0.002$ $[-0.13, +0.14]$
quantile transport	m4	-0.007 $[-0.043, +0.030]$	0.096	54	0.83	$+0.029$ $[-0.12, +0.18]$
MiMiC full-space affine	m4	$+0.013$ $[-0.037, +0.063]$	0.098	43	0.83	$+0.249$ $[+0.09, +0.40]$
sequential-gate ablation	m4	$+0.003$ $[-0.027, +0.037]$	0.098	49	0.93	$+0.074$ $[-0.03, +0.19]$
online_d0p05_q50_L16_t16_ablate	m4	$+0.017$ $[-0.023, +0.057]$	0.124	53	0.79	-0.011 $[-0.15, +0.13]$
online_d0p05_q50_L16_t64_ablate	m4	-0.007 $[-0.027, +0.013]$	0.075	47	0.93	$+0.054$ $[-0.05, +0.16]$
probegate-all_q60_ablate	m4	$+0.013$ $[-0.023, +0.053]$	0.105	42	0.74	$+0.117$ $[-0.04, +0.29]$
probegate-all_q60_clamp_q30	m4	$+0.027$ $[-0.010, +0.063]$	0.109	46	0.78	$+0.076$ $[-0.08, +0.24]$
probegate-all_q60_otq_cal	m4	-0.013 $[-0.043, +0.017]$	0.088	48	0.92	$+0.061$ $[-0.05, +0.19]$
probegate-all_q75_ablate	m4	$+0.020$ $[-0.013, +0.057]$	0.107	44	0.87	$+0.188$ $[+0.04, +0.34]$
probegate-all_q75_clamp_q30	m4	$+0.033$ $[+0.000, +0.070]$	0.074	40	0.86	$+0.195$ $[+0.05, +0.35]$
probegate-all_q75_otq_cal	m4	-0.003 $[-0.030, +0.027]$	0.089	51	0.89	$+0.035$ $[-0.08, +0.16]$
prompting	m4	-0.017 $[-0.073, +0.043]$	0.096	58	0.70	$+0.097$ $[-0.08, +0.26]$
LDA-gated ablation	m4	$+0.020$ $[-0.013, +0.053]$	0.079	38	0.83	$+0.209$ $[+0.04, +0.37]$
sweep_lda-allq50_clamp_q30	m4	$+0.030$ $[-0.003, +0.067]$	0.093	44	0.89	$+0.175$ $[+0.03, +0.33]$
sweep_lda-allq50_clamp_q50	m4	$+0.013$ $[-0.020, +0.047]$	0.100	45	0.95	$+0.187$ $[+0.06, +0.33]$
gated ablation	m4	$+0.033$ $[-0.020, +0.090]$	0.101	35	0.72	$+0.304$ $[+0.13, +0.47]$
sweep_ocwcr-crq50_clamp_q30	m4	$+0.007$ $[-0.040, +0.053]$	0.078	48	0.84	$+0.222$ $[+0.07, +0.38]$
gated clamp q_{50} (sweep)	m4	$+0.017$ $[-0.033, +0.067]$	0.081	47	0.86	$+0.195$ $[+0.04, +0.35]$
Linear-AcT (tuned)	m4	$+0.003$ $[-0.037, +0.043]$	0.090	48	0.88	$+0.182$ $[+0.03, +0.33]$
additive CAA (tuned)	m4	$+0.000$ $[-0.057, +0.063]$	0.079	40	0.59	$+0.232$ $[+0.05, +0.41]$
CAST-style (tuned)	m4	$+0.007$ $[-0.037, +0.053]$	0.076	33	0.76	$+0.123$ $[-0.04, +0.29]$
gated full-rank transport (tuned, ours)	m4	$+0.027$ $[-0.010, +0.063]$	0.090	46	0.92	$+0.181$ $[+0.05, +0.32]$
MiMiC (tuned)	m4	$+0.013$ $[-0.040, +0.070]$	0.110	56	0.75	$+0.070$ $[-0.09, +0.23]$
sequential gate (tuned, ours)	m4	-0.007 $[-0.037, +0.023]$	0.114	49	0.88	$+0.062$ $[-0.06, +0.19]$
gated ablation (tuned, ours)	m4	$+0.027$ $[-0.020, +0.077]$	0.094	35	0.78	$+0.276$ $[+0.11, +0.45]$
gated k -dim transport (tuned, ours)	m4	-0.027 $[-0.063, +0.013]$	0.095	57	0.82	-0.064 $[-0.19, +0.06]$
tuned_ourspost	m4	-0.003 $[-0.020, +0.013]$	0.075	48	0.99	$+0.067$ $[-0.01, +0.15]$
score-proportional dose (tuned, ours)	m4	$+0.043$ $[-0.010, +0.100]$	0.091	33	0.66	$+0.218$ $[+0.04, +0.39]$
gated token-local clamp (tuned, ours)	m4	$+0.023$ $[-0.023, +0.070]$	0.084	39	0.76	$+0.263$ $[+0.10, +0.44]$
baseline	m2	—	0.375	124	—	—
Linear-AcT $\lambda=0.5$	m2	$+0.017$ $[-0.013, +0.047]$	0.360	121	0.94	$+0.054$ $[-0.01, +0.12]$
Linear-AcT $\lambda=1.0$	m2	$+0.007$ $[-0.027, +0.040]$	0.374	124	0.92	$+0.052$ $[-0.02, +0.12]$
CAST-style gate+add	m2	-0.110 $[-0.167, -0.050]$	0.227	89	0.60	$+0.090$ $[-0.03, +0.20]$
ablation	m2	$+0.013$ $[-0.037, +0.063]$	0.344	120	0.84	$+0.106$ $[+0.01, +0.20]$
additive -0.75	m2	-0.193 $[-0.260, -0.127]$	0.219	98	0.38	-0.051 $[-0.16, +0.07]$
clamp q_{30}	m2	$+0.010$ $[-0.040, +0.057]$	0.354	122	0.85	$+0.104$ $[+0.01, +0.20]$

condition	r/o	Δacc [CI]	ECE	OCW	cr_keep	Sel [CI]
clamp q_{50}	m2	+0.007 [-0.037, +0.050]	0.368	125	0.88	+0.059 [-0.02, +0.14]
clamp q_{70}	m2	+0.003 [-0.043, +0.047]	0.380	126	0.87	+0.063 [-0.02, +0.15]
gated clamp q_{50}	m2	+0.023 [-0.020, +0.067]	0.355	119	0.90	+0.101 [+0.02, +0.18]
gated OT q_{50}	m2	+0.000 [-0.027, +0.027]	0.380	127	0.95	+0.009 [-0.04, +0.06]
gated clamp q_{70}	m2	+0.030 [-0.007, +0.067]	0.348	116	0.95	+0.116 [+0.04, +0.19]
gated OT q_{70}	m2	+0.010 [-0.007, +0.027]	0.370	121	0.99	+0.045 [+0.00, +0.09]
Gaussian-OT cal.	m2	-0.003 [-0.030, +0.023]	0.380	128	0.95	-0.005 [-0.05, +0.04]
quantile-OT cal.	m2	+0.010 [-0.017, +0.040]	0.368	125	0.95	+0.035 [-0.02, +0.10]
quantile transport	m2	+0.017 [-0.010, +0.047]	0.370	122	0.96	+0.055 [-0.00, +0.12]
MiMiC full-space affine	m2	+0.020 [-0.023, +0.063]	0.360	120	0.88	+0.091 [+0.01, +0.17]
sequential-gate ablation	m2	+0.007 [-0.010, +0.027]	0.363	123	0.98	+0.017 [-0.02, +0.06]
sequential-gate clamp q_{50}	m2	+0.000 [-0.017, +0.017]	0.377	124	0.98	+0.006 [-0.03, +0.04]
online_d0p05_q50.L16.t16.ablate	m2	+0.013 [-0.010, +0.040]	0.357	120	0.96	+0.053 [-0.00, +0.11]
online_d0p05_q50.L16.t64.ablate	m2	+0.010 [-0.007, +0.027]	0.369	121	0.99	+0.029 [-0.01, +0.07]
proagate-all_q60.ablate	m2	+0.000 [-0.033, +0.030]	0.367	122	0.93	+0.048 [-0.02, +0.12]
proagate-all_q60.clamp.q30	m2	-0.013 [-0.047, +0.020]	0.383	127	0.92	+0.020 [-0.04, +0.09]
proagate-all_q60.otq.cal	m2	+0.013 [-0.007, +0.033]	0.369	120	0.98	+0.047 [+0.00, +0.10]
proagate-all_q75.ablate	m2	+0.000 [-0.030, +0.030]	0.365	123	0.93	+0.018 [-0.04, +0.08]
proagate-all_q75.clamp.q30	m2	-0.007 [-0.037, +0.023]	0.378	126	0.92	+0.012 [-0.05, +0.08]
proagate-all_q75.otq.cal	m2	+0.007 [-0.017, +0.030]	0.367	123	0.97	+0.027 [-0.02, +0.07]
prompting	m2	+0.020 [-0.030, +0.073]	0.344	119	0.85	+0.110 [+0.01, +0.20]
LDA-gated ablation	m2	-0.007 [-0.033, +0.020]	0.365	125	0.94	+0.005 [-0.05, +0.06]
sweep_lda-allq50.clamp.q30	m2	+0.003 [-0.023, +0.030]	0.366	120	0.95	+0.043 [-0.02, +0.11]
sweep_lda-allq50.clamp.q50	m2	+0.003 [-0.020, +0.030]	0.365	123	0.96	+0.023 [-0.03, +0.08]
gated ablation	m2	+0.030 [-0.013, +0.073]	0.337	115	0.89	+0.119 [+0.03, +0.21]
sweep_ocwcr-crq50.clamp.q30	m2	+0.020 [-0.020, +0.063]	0.347	117	0.91	+0.129 [+0.05, +0.21]
gated clamp q_{50} (sweep)	m2	+0.023 [-0.020, +0.067]	0.355	119	0.90	+0.101 [+0.02, +0.18]
Linear-Act (tuned)	m2	+0.007 [-0.027, +0.040]	0.374	124	0.92	+0.052 [-0.02, +0.12]
additive CAA (tuned)	m2	+0.007 [-0.043, +0.053]	0.330	117	0.83	+0.108 [+0.02, +0.20]
CAST-style (tuned)	m2	+0.007 [-0.033, +0.047]	0.296	103	0.87	+0.152 [+0.06, +0.25]
gated full-rank transport (tuned, ours)	m2	+0.000 [-0.030, +0.030]	0.369	123	0.92	+0.036 [-0.03, +0.11]
MiMiC (tuned)	m2	+0.030 [-0.013, +0.073]	0.352	117	0.91	+0.123 [+0.04, +0.21]
sequential gate (tuned, ours)	m2	+0.007 [-0.020, +0.033]	0.365	122	0.95	+0.033 [-0.02, +0.09]
gated ablation (tuned, ours)	m2	+0.003 [-0.033, +0.043]	0.357	121	0.90	+0.053 [-0.02, +0.13]
gated k -dim transport (tuned, ours)	m2	-0.007 [-0.033, +0.017]	0.379	125	0.95	+0.011 [-0.04, +0.07]
tuned_ourspost	m2	-0.003 [-0.017, +0.007]	0.374	125	0.99	-0.004 [-0.03, +0.02]
score-proportional dose (tuned, ours)	m2	-0.013 [-0.057, +0.033]	0.371	126	0.85	+0.046 [-0.04, +0.13]
gated token-local clamp (tuned, ours)	m2	-0.003 [-0.040, +0.030]	0.360	122	0.90	+0.053 [-0.02, +0.13]

Table 15: ARC-Challenge (transfer): all conditions, both readouts.

condition	r/o	Δacc [CI]	ECE	OCW	cr_keep	Sel [CI]
baseline	m4	—	0.113	36	—	—
Linear-Act $\lambda=0.5$	m4	+0.010 [-0.023, +0.040]	0.112	39	0.95	+0.086 [-0.02, +0.22]
Linear-Act $\lambda=1.0$	m4	+0.003 [-0.033, +0.040]	0.097	41	0.90	+0.095 [-0.03, +0.25]
CAST-style gate+add	m4	-0.090 [-0.137, -0.047]	0.128	15	0.60	+0.179 [+0.00, +0.36]
ablation	m4	+0.020 [-0.033, +0.070]	0.110	38	0.79	+0.239 [+0.07, +0.42]
additive -0.75	m4	-0.147 [-0.207, -0.087]	0.239	0	0.01	+0.007 [+0.00, +0.02]
clamp q_{30}	m4	+0.017 [-0.037, +0.070]	0.090	35	0.80	+0.329 [+0.14, +0.50]
clamp q_{50}	m4	+0.027 [-0.027, +0.077]	0.127	37	0.84	+0.285 [+0.12, +0.46]
clamp q_{70}	m4	+0.010 [-0.033, +0.053]	0.080	41	0.82	+0.182 [+0.01, +0.35]
gated clamp q_{50}	m4	+0.003 [-0.037, +0.040]	0.110	36	0.90	+0.178 [+0.03, +0.33]
gated OT q_{50}	m4	-0.010 [-0.033, +0.013]	0.137	35	0.97	+0.057 [-0.03, +0.16]
gated clamp q_{70}	m4	+0.030 [+0.003, +0.057]	0.139	30	0.95	+0.197 [+0.06, +0.34]
gated OT q_{70}	m4	+0.000 [-0.020, +0.020]	0.103	36	0.99	+0.070 [-0.01, +0.17]
Gaussian-OT cal.	m4	+0.013 [-0.017, +0.043]	0.099	33	0.96	+0.182 [+0.05, +0.33]
quantile-OT cal.	m4	+0.017 [-0.017, +0.047]	0.111	37	0.97	+0.134 [+0.02, +0.27]
quantile transport	m4	+0.037 [+0.003, +0.073]	0.101	36	0.96	+0.210 [+0.07, +0.36]
MiMiC full-space affine	m4	-0.017 [-0.070, +0.033]	0.112	57	0.83	+0.084 [-0.07, +0.25]
sequential-gate ablation	m4	+0.003 [-0.010, +0.017]	0.123	35	0.99	+0.077 [+0.00, +0.18]
sequential-gate clamp q_{50}	m4	+0.000 [+0.000, +0.000]	0.104	38	1.00	+0.000 [+0.00, +0.00]
proagate-all_q60.ablate	m4	+0.013 [-0.030, +0.053]	0.086	36	0.89	+0.193 [+0.04, +0.36]
proagate-all_q60.clamp.q30	m4	+0.010 [-0.030, +0.050]	0.076	32	0.91	+0.241 [+0.08, +0.40]
proagate-all_q60.otq.cal	m4	+0.007 [-0.017, +0.027]	0.087	40	0.96	+0.044 [-0.04, +0.15]
proagate-all_q75.ablate	m4	+0.017 [-0.020, +0.050]	0.111	38	0.92	+0.198 [+0.05, +0.35]
proagate-all_q75.clamp.q30	m4	+0.020 [-0.017, +0.060]	0.093	31	0.95	+0.253 [+0.10, +0.41]
proagate-all_q75.otq.cal	m4	+0.007 [-0.020, +0.033]	0.087	37	0.97	+0.050 [-0.04, +0.15]
prompting	m4	+0.007 [-0.047, +0.057]	0.107	52	0.79	+0.267 [+0.08, +0.45]
LDA-gated ablation	m4	+0.010 [-0.020, +0.037]	0.116	36	0.95	+0.197 [+0.05, +0.35]
sweep_lda-allq50.clamp.q30	m4	+0.040 [+0.010, +0.070]	0.149	33	0.93	+0.128 [-0.00, +0.28]
sweep_lda-allq50.clamp.q50	m4	+0.027 [-0.000, +0.057]	0.144	37	0.95	+0.092 [-0.01, +0.22]
gated ablation	m4	-0.010 [-0.050, +0.027]	0.085	39	0.89	+0.221 [+0.06, +0.39]
sweep_ocwcr-crq50.clamp.q30	m4	+0.003 [-0.037, +0.040]	0.111	40	0.89	+0.200 [+0.04, +0.36]
gated clamp q_{50} (sweep)	m4	+0.003 [-0.037, +0.040]	0.110	36	0.90	+0.178 [+0.03, +0.33]
Linear-Act (tuned)	m4	+0.003 [-0.033, +0.040]	0.097	41	0.90	+0.095 [-0.03, +0.25]
additive CAA (tuned)	m4	+0.003 [-0.053, +0.060]	0.084	37	0.80	+0.385 [+0.21, +0.55]
CAST-style (tuned)	m4	+0.007 [-0.020, +0.033]	0.136	32	0.93	+0.184 [+0.04, +0.34]

condition	r/o	Δacc [CI]	ECE	OCW	cr_keep	Sel [CI]
gated full-rank transport (tuned, ours)	m4	+0.003 [-0.037, +0.043]	0.105	49	0.91	+0.136 [-0.00, +0.29]
MiMiC (tuned)	m4	-0.013 [-0.060, +0.033]	0.092	53	0.90	+0.151 [+0.01, +0.32]
sequential gate (tuned, ours)	m4	+0.020 [-0.007, +0.050]	0.109	36	0.97	+0.022 [-0.05, +0.11]
gated ablation (tuned, ours)	m4	+0.003 [-0.037, +0.043]	0.092	35	0.90	+0.290 [+0.13, +0.46]
gated k -dim transport (tuned, ours)	m4	+0.007 [-0.020, +0.033]	0.109	37	0.97	+0.078 [-0.02, +0.20]
tuned.ourspost	m4	+0.000 [+0.000, +0.000]	0.118	36	1.00	+0.000 [+0.00, +0.00]
score-proportional dose (tuned, ours)	m4	-0.003 [-0.057, +0.047]	0.084	37	0.81	+0.342 [+0.17, +0.51]
gated token-local clamp (tuned, ours)	m4	+0.043 [+0.003, +0.083]	0.129	37	0.87	+0.291 [+0.12, +0.46]
baseline	m2	—	0.242	78	—	—
Linear-AcT $\lambda=0.5$	m2	+0.007 [-0.013, +0.027]	0.230	73	0.98	+0.085 [+0.02, +0.16]
Linear-AcT $\lambda=1.0$	m2	+0.007 [-0.017, +0.030]	0.233	75	0.97	+0.075 [+0.01, +0.16]
CAST-style gate+add	m2	-0.113 [-0.160, -0.067]	0.192	89	0.74	-0.014 [-0.12, +0.10]
ablation	m2	+0.000 [-0.047, +0.043]	0.222	74	0.89	+0.233 [+0.12, +0.35]
additive -0.75	m2	-0.213 [-0.273, -0.153]	0.143	94	0.55	+0.014 [-0.11, +0.15]
clamp q_{30}	m2	-0.003 [-0.047, +0.040]	0.240	78	0.88	+0.177 [+0.07, +0.29]
clamp q_{50}	m2	-0.013 [-0.053, +0.023]	0.249	81	0.91	+0.106 [+0.01, +0.21]
clamp q_{70}	m2	+0.003 [-0.030, +0.037]	0.229	77	0.95	+0.112 [+0.02, +0.21]
gated clamp q_{50}	m2	-0.023 [-0.057, +0.003]	0.257	84	0.94	+0.026 [-0.04, +0.10]
gated OT q_{50}	m2	+0.003 [-0.017, +0.023]	0.235	76	0.98	+0.059 [+0.00, +0.13]
gated clamp q_{70}	m2	+0.000 [-0.020, +0.020]	0.237	76	0.98	+0.054 [-0.00, +0.12]
gated OT q_{70}	m2	-0.017 [-0.037, +0.000]	0.256	82	0.97	-0.002 [-0.04, +0.05]
Gaussian-OT cal.	m2	-0.007 [-0.030, +0.013]	0.241	78	0.97	+0.058 [-0.00, +0.13]
quantile-OT cal.	m2	+0.003 [-0.020, +0.023]	0.237	76	0.98	+0.067 [+0.01, +0.14]
quantile transport	m2	-0.010 [-0.030, +0.010]	0.242	80	0.97	+0.032 [-0.02, +0.09]
MiMiC full-space affine	m2	+0.027 [-0.010, +0.060]	0.217	68	0.95	+0.215 [+0.11, +0.33]
sequential-gate ablation	m2	+0.000 [-0.010, +0.010]	0.241	76	1.00	+0.034 [-0.00, +0.08]
sequential-gate clamp q_{50}	m2	+0.000 [-0.010, +0.010]	0.242	77	1.00	+0.021 [-0.01, +0.06]
probegate-all.q60.ablate	m2	+0.007 [-0.030, +0.040]	0.228	74	0.94	+0.155 [+0.06, +0.26]
probegate-all.q60.clamp.q30	m2	-0.007 [-0.037, +0.027]	0.234	78	0.93	+0.099 [+0.01, +0.18]
probegate-all.q60.otq.cal	m2	+0.010 [-0.003, +0.027]	0.231	74	1.00	+0.060 [+0.01, +0.12]
probegate-all.q75.ablate	m2	+0.027 [-0.007, +0.060]	0.219	70	0.95	+0.181 [+0.09, +0.29]
probegate-all.q75.clamp.q30	m2	+0.003 [-0.027, +0.037]	0.226	76	0.95	+0.112 [+0.02, +0.20]
probegate-all.q75.otq.cal	m2	+0.003 [-0.010, +0.017]	0.237	76	0.99	+0.042 [-0.00, +0.10]
prompting	m2	+0.010 [-0.027, +0.047]	0.231	75	0.93	+0.163 [+0.06, +0.27]
LDA-gated ablation	m2	+0.010 [-0.020, +0.037]	0.224	75	0.96	+0.105 [+0.03, +0.19]
sweep_lda-all.q50.clamp.q30	m2	-0.023 [-0.057, +0.007]	0.252	84	0.93	+0.035 [-0.04, +0.12]
sweep_lda-all.q50.clamp.q50	m2	-0.007 [-0.030, +0.017]	0.246	80	0.97	+0.032 [-0.02, +0.09]
gated ablation	m2	+0.000 [-0.033, +0.033]	0.229	77	0.93	+0.112 [+0.03, +0.20]
sweep_ocwcr-crq50.clamp.q30	m2	-0.033 [-0.067, +0.000]	0.267	85	0.91	+0.038 [-0.04, +0.12]
gated clamp q_{50} (sweep)	m2	-0.023 [-0.057, +0.003]	0.257	84	0.94	+0.026 [-0.04, +0.10]
Linear-AcT (tuned)	m2	+0.007 [-0.017, +0.030]	0.233	75	0.97	+0.075 [+0.01, +0.16]
additive CAA (tuned)	m2	+0.027 [-0.013, +0.067]	0.191	67	0.93	+0.266 [+0.16, +0.37]
CAST-style (tuned)	m2	-0.003 [-0.030, +0.023]	0.221	75	0.95	+0.083 [+0.01, +0.16]
gated full-rank transport (tuned, ours)	m2	+0.043 [+0.017, +0.070]	0.201	64	0.99	+0.204 [+0.11, +0.30]
MiMiC (tuned)	m2	+0.013 [-0.017, +0.043]	0.229	75	0.96	+0.126 [+0.04, +0.22]
sequential gate (tuned, ours)	m2	-0.010 [-0.033, +0.010]	0.248	79	0.97	+0.045 [-0.01, +0.12]
gated ablation (tuned, ours)	m2	+0.010 [-0.027, +0.043]	0.217	74	0.94	+0.167 [+0.07, +0.27]
gated k -dim transport (tuned, ours)	m2	+0.013 [-0.003, +0.033]	0.226	73	0.99	+0.081 [+0.02, +0.15]
tuned.ourspost	m2	+0.000 [+0.000, +0.000]	0.242	78	1.00	+0.000 [+0.00, +0.00]
score-proportional dose (tuned, ours)	m2	+0.020 [-0.027, +0.067]	0.207	71	0.91	+0.243 [+0.14, +0.36]
gated token-local clamp (tuned, ours)	m2	-0.003 [-0.043, +0.037]	0.243	79	0.90	+0.140 [+0.04, +0.24]

Table 16: GSM8K-MCQ (transfer): all conditions, both readouts. The OCW/CR poles are small (6 and 11 of 300 at baseline), the model is underconfident on this domain, selectivity intervals are wide, and the conditional operators act near-identity.

condition	r/o	Δacc [CI]	ECE	OCW	cr_keep	Sel [CI]
baseline	m4	—	0.258	6	—	—
Linear-AcT $\lambda=0.5$	m4	+0.043 [+0.000, +0.087]	0.297	5	0.64	-0.030 [-0.50, +0.56]
Linear-AcT $\lambda=1.0$	m4	+0.023 [-0.027, +0.070]	0.269	6	0.73	+0.227 [-0.31, +0.76]
CAST-style gate+add	m4	-0.240 [-0.313, -0.170]	0.055	0	0.00	+0.000 [+0.00, +0.00]
ablation	m4	-0.033 [-0.090, +0.023]	0.245	9	0.18	-0.151 [-0.67, +0.29]
additive -0.75	m4	-0.267 [-0.337, -0.200]	0.033	0	0.00	+0.000 [+0.00, +0.00]
clamp q_{30}	m4	-0.003 [-0.067, +0.057]	0.260	8	0.36	-0.136 [-0.67, +0.43]
clamp q_{50}	m4	+0.027 [-0.027, +0.077]	0.289	8	0.45	+0.121 [-0.46, +0.62]
clamp q_{70}	m4	+0.033 [-0.017, +0.087]	0.300	10	0.55	+0.045 [-0.50, +0.60]
gated clamp q_{50}	m4	+0.030 [-0.020, +0.080]	0.287	7	0.55	+0.212 [-0.36, +0.70]
gated OT q_{50}	m4	+0.040 [-0.007, +0.087]	0.297	5	0.55	+0.045 [-0.50, +0.62]
gated clamp q_{70}	m4	+0.017 [-0.030, +0.063]	0.279	7	0.55	+0.212 [-0.33, +0.71]
gated OT q_{70}	m4	+0.010 [-0.030, +0.047]	0.268	5	0.64	-0.030 [-0.50, +0.55]
Gaussian-OT cal.	m4	+0.020 [-0.027, +0.063]	0.280	5	0.73	+0.061 [-0.43, +0.62]
quantile-OT cal.	m4	+0.047 [+0.003, +0.093]	0.295	5	0.64	+0.136 [-0.42, +0.70]
quantile transport	m4	+0.003 [-0.037, +0.043]	0.257	5	0.73	+0.227 [-0.33, +0.80]
MiMiC full-space affine	m4	+0.063 [+0.007, +0.120]	0.281	11	0.45	-0.045 [-0.60, +0.50]
sequential-gate ablation	m4	+0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
probegate-all.q60.ablate	m4	+0.007 [-0.013, +0.027]	0.262	6	0.91	+0.076 [-0.25, +0.50]
probegate-all.q60.clamp.q30	m4	+0.000 [-0.017, +0.017]	0.256	8	1.00	+0.000 [+0.00, +0.00]

condition	r/o	Δacc [CI]	ECE	OCW	cr_keep	Sel [CI]
1296						
1297	proagate-all.q60.otq.cal	m4 +0.010 [-0.003, +0.023]	0.257	6	1.00	+0.167 [+0.00, +0.50]
1298	proagate-all.q75.ablate	m4 +0.007 [-0.007, +0.020]	0.262	6	0.91	+0.076 [-0.25, +0.50]
1299	proagate-all.q75.clamp.q30	m4 +0.003 [-0.007, +0.013]	0.262	6	1.00	+0.167 [+0.00, +0.50]
1300	proagate-all.q75.otq.cal	m4 +0.000 [-0.010, +0.010]	0.257	6	1.00	+0.167 [+0.00, +0.50]
1301	prompting	m4 -0.030 [-0.090, +0.033]	0.195	15	0.27	+0.106 [-0.36, +0.50]
1302	LDA-gated ablation	m4 +0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
1303	sweep_lda-allq50.clamp.q30	m4 +0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
1304	sweep_lda-allq50.clamp.q50	m4 +0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
1305	gated ablation	m4 -0.050 [-0.107, +0.010]	0.235	11	0.18	-0.318 [-0.86, +0.22]
1306	sweep_ocwcr-crq50.clamp.q30	m4 -0.007 [-0.067, +0.047]	0.267	10	0.36	-0.303 [-0.79, +0.25]
1307	gated clamp q_{50} (sweep)	m4 +0.030 [-0.020, +0.080]	0.287	7	0.55	+0.212 [-0.36, +0.70]
1308	Linear-Act (tuned)	m4 +0.023 [-0.027, +0.070]	0.269	6	0.73	+0.227 [-0.31, +0.76]
1309	additive CAA (tuned)	m4 -0.030 [-0.093, +0.027]	0.243	1	0.00	+0.000 [+0.00, +0.00]
1310	CAST-style (tuned)	m4 -0.063 [-0.120, -0.010]	0.218	7	0.27	+0.106 [-0.33, +0.50]
1311	gated full-rank transport (tuned, ours)	m4 +0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
1312	MiMiC (tuned)	m4 -0.013 [-0.070, +0.040]	0.233	11	0.55	+0.045 [-0.47, +0.59]
1313	sequential gate (tuned, ours)	m4 +0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
1314	gated ablation (tuned, ours)	m4 +0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
1315	gated k -dim transport (tuned, ours)	m4 +0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
1316	tuned_ourspost	m4 +0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
1317	score-proportional dose (tuned, ours)	m4 +0.003 [-0.007, +0.013]	0.259	6	1.00	+0.000 [+0.00, +0.00]
1318	gated token-local clamp (tuned, ours)	m4 +0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
1319	baseline	m2 —	0.216	87	—	—
1320	Linear-Act $\lambda=0.5$	m2 +0.013 [-0.017, +0.047]	0.204	85	0.94	+0.092 [+0.01, +0.18]
1321	Linear-Act $\lambda=1.0$	m2 +0.020 [-0.010, +0.053]	0.202	87	0.95	+0.121 [+0.04, +0.21]
1322	CAST-style gate+add	m2 -0.280 [-0.347, -0.217]	0.226	102	0.30	-0.107 [-0.23, +0.01]
1323	ablation	m2 +0.030 [-0.020, +0.077]	0.171	70	0.88	+0.335 [+0.22, +0.45]
1324	additive -0.75	m2 -0.290 [-0.360, -0.220]	0.218	105	0.26	-0.081 [-0.20, +0.03]
1325	clamp q_{30}	m2 +0.040 [+0.000, +0.073]	0.192	71	0.94	+0.283 [+0.18, +0.38]
1326	clamp q_{50}	m2 +0.013 [-0.033, +0.057]	0.213	74	0.89	+0.288 [+0.17, +0.40]
1327	clamp q_{70}	m2 +0.030 [-0.010, +0.070]	0.192	69	0.94	+0.283 [+0.18, +0.39]
1328	gated clamp q_{50}	m2 +0.030 [-0.010, +0.070]	0.199	78	0.91	+0.222 [+0.12, +0.33]
1329	gated OT q_{50}	m2 +0.023 [-0.013, +0.060]	0.191	76	0.94	+0.168 [+0.07, +0.26]
1330	gated clamp q_{70}	m2 +0.030 [-0.010, +0.067]	0.196	77	0.92	+0.181 [+0.09, +0.28]
1331	gated OT q_{70}	m2 +0.017 [-0.017, +0.050]	0.205	82	0.93	+0.128 [+0.04, +0.22]
1332	Gaussian-OT cal.	m2 -0.010 [-0.043, +0.027]	0.224	91	0.92	+0.055 [-0.02, +0.14]
1333	quantile-OT cal.	m2 +0.003 [-0.027, +0.033]	0.201	80	0.94	+0.138 [+0.05, +0.23]
1334	quantile transport	m2 +0.000 [-0.037, +0.033]	0.212	85	0.92	+0.078 [-0.00, +0.17]
1335	MiMiC full-space affine	m2 +0.030 [-0.020, +0.080]	0.194	77	0.87	+0.307 [+0.19, +0.42]
1336	sequential-gate ablation	m2 +0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
1337	proagate-all.q60.ablate	m2 -0.007 [-0.027, +0.010]	0.220	89	0.97	+0.003 [-0.04, +0.05]
1338	proagate-all.q60.clamp.q30	m2 +0.003 [-0.010, +0.017]	0.212	86	0.98	+0.019 [-0.02, +0.07]
1339	proagate-all.q60.otq.cal	m2 +0.003 [-0.013, +0.020]	0.216	86	0.98	+0.030 [-0.01, +0.08]
1340	proagate-all.q75.ablate	m2 -0.003 [-0.020, +0.010]	0.222	91	0.98	-0.004 [-0.03, +0.03]
1341	proagate-all.q75.clamp.q30	m2 +0.013 [+0.003, +0.027]	0.208	86	0.99	+0.006 [-0.01, +0.03]
1342	proagate-all.q75.otq.cal	m2 -0.010 [-0.023, +0.000]	0.224	90	0.98	-0.004 [-0.03, +0.03]
1343	prompting	m2 +0.010 [-0.043, +0.063]	0.226	88	0.83	+0.191 [+0.08, +0.31]
1344	LDA-gated ablation	m2 +0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
1345	sweep_lda-allq50.clamp.q30	m2 +0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
1346	sweep_lda-allq50.clamp.q50	m2 +0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
1347	gated ablation	m2 +0.013 [-0.040, +0.063]	0.190	76	0.84	+0.258 [+0.14, +0.37]
1348	sweep_ocwcr-crq50.clamp.q30	m2 +0.043 [+0.003, +0.083]	0.179	72	0.93	+0.295 [+0.19, +0.40]
1349	gated clamp q_{50} (sweep)	m2 +0.030 [-0.010, +0.070]	0.199	78	0.91	+0.222 [+0.12, +0.33]
1350	Linear-Act (tuned)	m2 +0.020 [-0.010, +0.053]	0.202	87	0.95	+0.121 [+0.04, +0.21]
1351	additive CAA (tuned)	m2 +0.023 [-0.030, +0.077]	0.133	69	0.80	+0.343 [+0.23, +0.46]
1352	CAST-style (tuned)	m2 -0.060 [-0.110, -0.013]	0.204	77	0.76	+0.182 [+0.05, +0.30]
1353	gated full-rank transport (tuned, ours)	m2 +0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
1354	MiMiC (tuned)	m2 -0.007 [-0.053, +0.040]	0.249	91	0.86	+0.147 [+0.04, +0.26]
1355	sequential gate (tuned, ours)	m2 +0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
1356	gated ablation (tuned, ours)	m2 +0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
1357	gated k -dim transport (tuned, ours)	m2 +0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
1358	tuned_ourspost	m2 +0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
1359	score-proportional dose (tuned, ours)	m2 +0.000 [-0.010, +0.010]	0.217	85	0.99	+0.013 [-0.02, +0.05]
1360	gated token-local clamp (tuned, ours)	m2 +0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]

K CAPABILITY OUTSIDE THE TARGET BEHAVIOR

We fit the operators on multiple-choice reasoning traces, so the question is what they do to generation that has nothing to do with the target behavior. Table 17 reports code synthesis (HumanEval pass@1, greedy, $n=164$), held-out language modelling (WikiText-103 perplexity, $n=300$ passages), and open-ended instruction following (200 AlpacaEval prompts, 384 new tokens): fluency as the perplexity the unsteered model assigns to the continuation, repetition as distinct-3, and length in words. The last column is the fraction of open-ended prompts on which the sequential gate fires at all; it fires on none of HumanEval, and WikiText is scored teacher-forced, which gives the gate no

method	HumanEval pass@1	WikiText ppl	open-ended (AlpacaEval prompts)			gate fires
			fluency ppl	distinct-3	win rate	
unsteered	0.396	23.4	3.00	0.953	—	—
additive —0.75 Panickssery et al. (2024)	0.128	47.3	3.36	0.480	0.043	—
additive, tuned dose Panickssery et al. (2024)	0.408	23.6	3.14	0.922	0.495	—
MiMiC Singh et al. (2024)	0.396	24.2	3.01	0.949	0.512	—
MiMiC, tuned layer Singh et al. (2024)	0.384	24.6	3.02	0.948	0.458	—
Linear-AcT Rodriguez et al. (2025a)	0.390	23.4	2.98	0.950	0.505	—
clamp q_{50} (ours)	0.390	23.1	2.97	0.946	0.507	—
quantile transport (ours)	0.396	23.3	2.99	0.951	0.492	—
online-gated ablation (ours)	0.396	n/a	3.01	0.946	0.477	0.680
gated full-rank transport (ours)	0.396	n/a	3.02	0.954	0.470	0.586

Table 17: **Capability away from the fitted behavior.** The published additive dose takes HumanEval pass@1 from .396 to .128 and makes 57.5% of open-ended generations degenerate; every projection-local operator and both gated operators sit at the unsteered value on every measurement available to them. The unconditional rows keep the action on for every token, so they bound what the gated operator can cost. We score WikiText teacher-forced, which gives a sequential gate no generated prefix to read, so that cell is not applicable. The win rate is order-consistent against the unsteered answer, judged by Qwen2.5-7B-Instruct in both presentation orders with ties counted as half, so 0.5 means indistinguishable.

generated prefix to read. We count a generation as degenerate when it fails to terminate or repeats a trigram cycle: 57.5% of them under the published additive dose, none under any operator of ours.

L DO STEERED ACTIVATIONS STAY ON THE MANIFOLD?

Transfer does not push activations off the manifold. We condition the edited coordinates on the untouched complement, which gives a statistic that is χ_k^2 under the null, so the fraction of tokens outside the 99% conditional ellipsoid has a value to compare against: unsteered traces sit at 0.9–2.9%. The published additive shift puts 100% of tokens outside it on all three domains and the equal-budget dose 40–70%, while clamping, quantile transport, MiMiC and Linear-AcT stay within a point or two of the unsteered level on domains their maps never saw (Table 18).

Geometry and behavior disagree, and both matter. The two diagnostics do not rank the operators the same way, so we report both. Measured by the KL the edit induces on the next-token distribution, the accepted locality criterion in this literature Arditi et al. (2024), the worst operator is a *random* direction at the same norm as the additive dose: 0.44 on MMLU against a 0.1 threshold, where the additive dose itself sits at 0.032 and the projection-local operators between 0.000 and 0.010. Yet that random direction is invisible to the conditional test, because it barely projects onto the behavioral coordinates the test watches. The conditional test asks whether the behavioral coordinates were moved to an implausible place and the KL asks whether the next token noticed. An operator has to pass both, and the transport maps do, ungated or gated. Directional ablation does not: it passes the KL test at 0.010 and fails the conditional one at 29.5%, which is the geometric form of the retention it spends in Table 2. The composed gated operator applies that action on the traces the gate fires on, so we do not claim a conditional-ellipsoid figure for it; what we report for it is the behavioral evidence of Appendix K.

We teacher-force the unsteered traces, apply each operator to the layer-14 residuals, and measure how far the result travels. Table 18 reports the relative displacement $\|h' - h\|/\|h\|$, the 99th percentile of the norm ratio $\|h'\|/\|h\|$, and the distance to the nearest natural activation in a bank of 4,096 unsteered residuals relative to the unsteered nearest-neighbour distance. We include the transfer domains ARC and GSM8K because that is where a map fitted on MMLU could be expected to explode.

We state one negative result here. A *full-space* Mahalanobis distance is useless here: an operator that edits a k -dimensional projection leaves $d - k$ coordinates untouched, which dilute the statistic until it moves by under 1% for every operator in the table, including the one that destroys generation. The same dilution has been reported independently Vogels et al. (2025). We therefore condition instead. Writing $h = (q, h_\perp)$ with $q = Vh$, the law of q given the 64 leading principal directions of the complement is Gaussian under the layer’s fitted moments, and its squared Mahalanobis distance is χ_k^2 under the null, so the natural readout is the fraction of tokens outside the 99% conditional ellipsoid, against an expected 1%. We also report the behavioral check the refusal literature uses

operator	MMLU				ARC				GSM8K			
	disp.	NN	out%	KL	disp.	NN	out%	KL	disp.	NN	out%	KL
unsteered	—	1.00	0.9	0.000	—	1.00	1.2	0.000	—	1.00	2.9	0.000
random direction, norm-matched	0.663	1.90	1.6	0.445	0.644	2.27	1.7	0.233	0.749	2.06	0.9	0.149
additive -0.75 Panickssery et al. (2024)	0.663	1.88	100.0	0.032	0.644	2.25	100.0	0.008	0.749	2.04	100.0	0.012
additive, tuned dose Panickssery et al. (2024)	0.221	1.22	69.9	0.005	0.215	1.33	40.3	0.001	—	—	—	—
MiMiC Singh et al. (2024)	0.082	1.06	1.3	0.001	0.074	1.08	1.8	0.000	0.097	1.06	4.5	0.001
Linear-AcT Rodriguez et al. (2025a)	0.012	1.01	1.0	0.000	0.011	1.01	1.5	0.000	0.012	1.01	3.5	0.000
clamp q_{50} (ours)	0.034	1.03	1.3	0.006	0.046	1.06	1.1	0.001	0.003	1.00	2.9	0.001
quantile transport (ours)	0.004	1.00	1.0	0.000	0.003	1.00	1.3	0.000	0.007	1.00	4.0	0.000
ablation Arditi et al. (2024)	0.144	1.15	29.5	0.010	0.178	1.28	28.4	0.001	0.062	1.05	10.1	0.001

Table 18: On-manifold diagnostics at layer 14. NN is the mean nearest-neighbour distance to a bank of natural activations relative to the unsteered value, so 1.00 means the edit lands as close to the model’s own states as the states it replaced; out% is the fraction of tokens outside the 99% conditional ellipsoid, against 1% expected; KL is the mean divergence of the next-token distribution, with 0.1 the accepted threshold Arditi et al. (2024).

method	passes	seconds	wall clock
unsteered	1	43.5	1.00×
additive -0.75 Panickssery et al. (2024)	1	45.2	1.04×
ablation Arditi et al. (2024)	1	44.6	1.02×
clamp q_{50} (ours)	1	41.4	0.95×
quantile transport (ours)	1	44.8	1.03×
MiMiC Singh et al. (2024)	1	42.9	0.98×
Linear-AcT Rodriguez et al. (2025a)	1	40.9	0.94×
gated ablation, post-hoc gate (ours) (223/300 regenerated)	2	85.8	1.97×
gated ablation, sequential gate (ours)	1	45.5	1.05×

Table 19: End-to-end wall clock, MMLU $n=300$, batch 64, one H100.

as a selection criterion, the mean KL between the unsteered and steered next-token distributions with 0.1 as the accepted threshold Arditi et al. (2024), measured against a norm-matched random direction so that the number is interpretable Turner et al. (2023). We claim only that a projection-local edit lands closer to the states the model produces by itself, not that it lands on them: steered states are provably unreachable from any discrete prompt Mishra et al. (2026).

M END-TO-END SERVING COST

Table 19 measures the whole 300-question workload on one H100, so the post-hoc gate is charged for its scoring pass and for regenerating what it flags.

N DEFINITIONS USED IN THE TEXT

Weights. 2B weights were served from the unsloth/gemma-2-2b-it mirror; all 288 tensors are bit-identical to the official google/gemma-2-2b-it release.

Confidence readout prompts. The M4 yes-probability readout scores each option with "Answer whether the given answer is the right answer... Question: {q} Answer: {option}", one forward per option, reasoning in <think>, then YES/NO, and the four $P(\text{YES})$ values are renormalised.

Hedge markers. The lexicon counted in Sec. 5 and Appendix F is fixed in advance and case-insensitive: *maybe, perhaps, might, possibly, however, wait, actually, hmm, unsure, unclear, guess, seems, probably, could, uncertain, confused*. The matching certainty lexicon is *clearly, obviously, definitely, certainly, surely, undoubtedly, evidently, must, certain, sure*. A trace’s hedge count is the number of token matches; rates are per trace.

Calibrated zone. The green ellipse in Fig. 1 is the 2σ contour of the empirical Gaussian fitted to the trace-mean projections of the *calibrated* extraction-split states (CR and nonconfident-wrong), i.e. the states the transport target is built from. It is a visual reference, not a decision boundary.

Norm-matched spherical action. The comparison action in Table 5 is a spherical steering operator: activations are rotated along the fixed-azimuth shortest geodesic towards a unit prototype at constant

norm, with the decision taken by an antipodal von Mises–Fisher confidence at concentration $\kappa=20$. Its nine-cell grid ($\alpha \in \{0.6, 0.9, 1\}$, $\beta \in \{0.6, 0.9, 0.99\}$) is selected on the observer split, and the reported comparison uses the selected cell, so the two conditions differ in the direction and the decision rule and not in how far they move activations. The five *isotropic* controls replace the detector direction with a random unit direction at exactly the measured update norm; the *unprompted* control refits the direction at last-token positions without the evaluation instruction. All are input-independent given the fit.

Condition names. The exhaustive grids in Appendix J carry the run identifiers; the main text uses short names. `m4_conf_*` is an action on the confidence axis; `sweep_ocwcr-crq50_ablate` is gated ablation (gate fitted on the OCW-versus-CR contrast, threshold at the CR median); `sweep_lda-allq50_ablate` is LDA-gated ablation (the two-dimensional LDA head of Sec. 3, threshold at the median of all extraction scores); `online_dp05-q50_L16_ablate` is its sequential single-pass form at $\delta=0.05$; `probegate-*` gates on a full- d logistic probe; `cast_gate-cr-q50_add-0.75` is the CAST-style reimplementation; `act_l05/act_l10` are Linear-AcT at $\lambda=0.5/1.0$; `mimic_full` is MiMiC; `subspace-BW` is ungated subspace Bures–Wasserstein transport and `NP-gate $\times$ BW` its gated form (`subbw` and `npbw` in the run identifiers, and gated k -dim transport and gated full-rank transport in the tables); `tuned_ourspost` is the posterior-weighted dose of Appendix I; `self_qXX` marks a threshold re-anchored, label-free, to the target domain’s own score quantile; and `-think` marks steering confined to the `<think>` phase.